Who bears the AI shock? The distributional and fiscal incidence in the UK*

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Abstract

I study how the simulated household and fiscal effects of an AI labour-market shock in the United Kingdom vary with its incidence. Linking occupation-level AI exposure to workers in the 2024–25 Family Resources Survey, I use PolicyEngine UK to simulate employment, wage and capital-income shocks across 66 shock-size combinations and five displacement-incidence families that hold aggregate displacement fixed. In the central 7 per cent displacement calibration, the mean Exchequer cost is £17.7 billion per year and absolute before-housing-costs poverty rises by 1.87 percentage points. Incidence matters fiscally: the same aggregate shock costs £13.9 billion under uniform incidence and up to £32.8 billion under a top-loaded stress test, while poverty varies less. The Gini rises in every displacement scenario examined; shifting the same earnings loss from job losses to occupation-level wage cuts instead raises the fiscal cost, leaves poverty nearly unchanged and slightly lowers the Gini. Both incidence and the adjustment margin matter for household outcomes and the public finances.

Keywords: artificial intelligence, labour market, microsimulation, poverty, inequality, public finances

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1 Introduction

If artificial intelligence displaces a substantial share of UK employment, who bears the resulting loss: displaced workers, lower-income households or the Exchequer? The answer depends not only on the size of the shock but also on how it is distributed. The UK's services-intensive economy has substantial employment in professional, financial, administrative and managerial occupations that score highly on common measures of AI exposure (Felten et al., 2021; Eloundou et al., 2024; DSIT and DfE, 2023; Henseke et al., 2025). Its tax-benefit system combines means-tested support with substantial taxation of labour income, so the household and fiscal effects depend on which workers lose earnings.

Exposure indices do not, by themselves, establish that displacement will be concentrated towards the top of the earnings distribution. Evidence on AI's labour-market incidence remains unsettled. Exposure-based scenarios assign greater risk to highly paid cognitive work (Felten et al., 2021; Eloundou et al., 2024); an alternative account emphasises compression of expertise premia, with less experienced workers gaining access to tasks previously performed by more highly paid specialists (Autor, 2024); and several early empirical studies report weaker employment outcomes for junior workers in exposed occupations or firms (Klein Teeselink, 2025; Hosseini Maasoum and Lichtinger, 2026a; Brynjolfsson et al., 2025a). These are distinct hypotheses about incidence rather than refinements of a single estimate. They imply different household losses, benefit responses and reductions in tax receipts.

I treat incidence as an explicit dimension of the scenario design. Using PolicyEngine UK and processed Family Resources Survey 2024–25 microdata, I assign occupation-level AI exposure to workers and compare five displacement allocations: exposure-proportional, junior-concentrated, expertise-compression, uniform, and a Klein-anchored top-loaded stress test. The simulation adapts the Irish tax-benefit framework of Doorley et al. (2026), including its 7 per cent employment-loss and 2.6 per cent survivor-wage conventions, while extending the design by varying incidence explicitly. I also simulate 66 combinations of displacement rates and wage uplifts, together with a separate wage-adjustment family in which author-imposed occupation-level cuts replace job losses. The code and aggregate artefacts are public, but exact regeneration requires licensed FRS microdata. Plain survey weights and the absence of the official SPI top-income adjustment limit representativeness at the top of the distribution.

Two findings organise the paper. First, measured inequality rises in every displacement calibration examined: across all 66 shock-size cells, the Gini change ranges from 0.06 to 1.61 percentage points; across the five central incidence families, the 50-draw means range from 0.97 to 1.16 points. This is a conditional result of the simulated displacement, survivor-wage and capital-income channels, not a claim about every possible AI adjustment. In the exposure-proportional case, selected higher-income households lose earnings while surviving workers receive wage gains and capital-income gains remain concentrated near the top. Lower tax liabilities cushion some losses but do not prevent the simulated disposable-income distribution from widening.

Second, incidence changes the balance between fiscal and poverty effects. Holding the weighted displacement quota and central parameters fixed, the 50-draw mean Exchequer cost ranges from £13.9 billion under uniform incidence to £20.6 billion under expertise compression and £32.8 billion under the Klein-anchored stress test. Across the four stylised families, absolute before-housing-costs poverty rises by 1.85 to 1.96 percentage points; because the families share assignment seeds, paired draw-level comparisons show the expertise-compression poverty effect exceeding the exposure-proportional one (by +0.09 points, paired 95 per cent interval $[+0.03, +0.15]$), while the uniform–exposure poverty difference (-0.00 points, $[-0.06, +0.06]$)

and the junior–exposure fiscal-cost difference are statistically indistinguishable. The adjustment margin also matters. In the paired seed-0 comparison, moving the same 7.77 per cent gross earnings loss from wage cuts to job loss lowers the simulated fiscal cost from £26.6 billion to £18.2 billion, while raising poverty from 0.10 to 1.81 points and the Gini change from -0.44 to $+1.04$ points. These comparisons quantify the consequences of the specified incidence and adjustment rules; they do not identify which pattern will occur.

The remainder of the paper reviews the literature, sets out the scenario and microsimulation methods, reports the distributional and fiscal results, examines three stylised policy responses, and concludes with limitations and implications.

2 Related literature

My study connects two literatures: competing accounts of where AI-related labour-market adjustment may fall, and microsimulation studies that trace labour-market shocks through tax-benefit systems to household disposable income. I represent the main incidence views as separate scenarios and compare them within the same microsimulation framework.

2.1 Measuring exposure to artificial intelligence

The modern exposure literature descends from the occupation-level automation probabilities of Frey and Osborne (2017), whose estimate that 47 per cent of US employment was at high risk of computerisation set the template of scoring occupations against machine capabilities. Subsequent work replaced whole-occupation probabilities with task- and ability-level measurement: Webb (2019) matches patent text to job-task descriptions; Felten et al. (2021) link progress in ten AI application areas to the O*NET ability requirements of each occupation, producing the AI Occupational Exposure index I build on; and Tolan et al. (2021) connect AI benchmark results to cognitive abilities and tasks. The arrival of large language models prompted a further generation of measures targeted at generative AI specifically: the task-level rubric of Eloundou et al. (2024), the ILO's global assessment of jobs at risk of transformation rather than elimination (ILO, 2025), the UK government's occupational assessment (DSIT and DfE, 2023), and the GAISI index of Henseke et al. (2025). What no index can supply is incidence: an index records what AI *could* do to an occupation's task bundle, and translating it into who actually loses work requires assumptions the indices themselves cannot supply. Hosseini Maasoum and Lichtinger (2026b) show that exposure is an incomplete statistic even for the supply side: their measure of the potential supply shift—the expansion of the qualified worker pool as generative AI lowers expertise requirements—varies widely across occupations with identical exposure scores, and they publish it as an alternative public occupation-level measure, which I use as a wage-cut gradient in the wage-margin family of Section 4.6. The complementarity adjustment of Pizzinelli et al. (2023), which I implement in Section 3.1, is one such assumption; the incidence families of Section 3.4 treat the remainder of that translation as an explicit scenario choice.

2.2 The incidence debate

Exposure identifies which occupations AI can reach; it does not settle who bears any resulting adjustment. I distinguish three incidence views—exposure-proportional, junior-concentrated and expertise-compression—before reviewing evidence that motivates smaller or differently distributed shocks.

The exposure school. A common scenario interpretation uses exposure as a proxy for incidence: occupations with higher scores are assigned greater displacement risk and, under that assumption, the indices point towards the top of the labour market. They consistently show exposure rising with education and earnings—professional, managerial and administrative occupations score highest, a pattern that recurs across European labour markets (Williamson et al., 2025) and, for the UK specifically, in the government's official assessment (DSIT and DfE, 2023) and in the GAISI index of Henseke et al. (2025). On this view AI breaks with the routine-biased pattern of earlier waves, in which computerisation hollowed out codifiable mid-skill work (Autor et al., 2003; Goos and Manning, 2007; Goos et al., 2014; Acemoglu and Restrepo, 2022), and instead aims at the top of the wage distribution. The IMF strand adds an important qualification: high exposure need not mean displacement, because the same workers may enjoy the largest complementarity gains. Cazzaniga et al. (2024) formalise this ambiguity, and Pizzinelli et al. (2023) document cross-country variation in the exposure–complementarity mix; Rockall et al. (2025) apply the framework to the UK and conclude that AI could *reduce* wage inequality if displacement falls disproportionately on high earners—unless complementarity among those same workers offsets it.

The closest methodological precedent is Doorley et al. (2026). They combine C-AIOE with Irish SILC microdata and the SWITCH tax-benefit model, and their central scenario uses the same 7 per cent employment loss, 2.6 per cent survivor-wage gain and 0.4 percentage-point increase in the return to capital adopted here. Their study provides an Irish exposure-proportional benchmark, alongside uniform-incidence and alternative-index sensitivities. My analysis transfers those allocation mechanics to UK data and extends the design by varying incidence and adjustment margin and by adding UK policy counterfactuals. Differences in country, microdata, occupational resolution and tax-benefit model mean that the numerical results are comparable conditional simulations, not a replication or independent validation. Earlier work by Doorley et al. (2023) carries automation shocks through tax-benefit models across European countries and finds that progressive taxes and means-tested transfers mute the pass-through of wider wage dispersion to household inequality.

The junior-concentrated evidence. Several early empirical studies report a junior gradient within exposed occupations. Klein Teeselink (2025) link generative-AI exposure to UK firm-level records and find employment reductions of around 4.5 per cent at more exposed firms, concentrated in junior positions; Hosseini Maasoum and Lichtinger (2026a) document, in US resume and posting data, junior employment falling by around 9 per cent within six quarters of firm AI adoption while senior employment is unaffected; and Brynjolfsson et al. (2025a) find a 16 per cent relative employment decline among workers aged 22–25 in the most exposed US occupations since late 2022, with older workers largely spared. If incidence is graded by seniority as much as by skill, the distributional consequences differ from a classic skill-biased shock, because junior workers are spread across the household income distribution and interact differently with means-tested benefits.

The expertise-compression school. A competing view, associated with Autor (2024), holds that AI's distinctive property is not that it targets exposed occupations but that it compresses the returns to accumulated expertise. By supplying decision-relevant knowledge on demand, AI lets less experienced and less credentialed workers perform tasks previously reserved for elite professionals—an opportunity to rebuild middle-class work, but a threat to the wage premia of the experienced workers whose scarcity those premia reflect. Experimental evidence is consistent with the mechanism: generative AI raises output most for the least experienced workers, compressing within-occupation productivity gaps in customer support (Brynjolfsson et al., 2025b) and professional writing (Noy and Zhang, 2023). Hosseini Maasoum and Lichtinger (2026b) formalise and quantify this mechanism: using LLM ratings of O*NET tasks they construct occupation-level potential supply shifts—the expansion of the qualified worker pool as generative AI lowers expertise requirements—and find in a calibrated general-equilibrium model that entry-barrier erosion compresses between-occupation wage inequality through wage adjustment rather than displacement, with losses concentrated in mid-to-upper-expertise scarcity premia and the extreme top largely insulated. On this view the shock lands on senior premia rather than junior heads—nearly the mirror image of the junior-concentrated evidence above, which is precisely why incidence needs to be varied rather than assumed.

Evidence motivating smaller or differently distributed shocks. Two further strands inform the scenario range without defining an additional family. The first concerns its *direction*: by making prediction cheap, AI raises the value of the human complements to prediction—judgement, decision-making and responsibility (Agrawal et al., 2018)—so that workers whose roles centre on decisions rather than predictable task execution may gain, pushing incidence towards prediction-like tasks and away from the senior decision-makers on whom the expertise-compression view lands it. The second concerns its *magnitude*: realised effects may simply be small. Danish population-scale evidence finds minimal effects of AI chatbots on earnings and hours in the first years of diffusion (Humlum and Vestergaard, 2025), US vacancy data show exposed establishments reorganising hiring without detectable aggregate employment effects (Acemoglu et al., 2022), and Acemoglu (2025) derives modest macroeconomic gains from task-level foundations. Acemoglu and Restrepo (2022) show that even historically transformative automation delivered only a cumulative 3.4 per cent TFP gain over 1980–2016 while reshaping the entire wage structure—large distributional effects do not require large aggregate ones. US evidence on publicly funded job training likewise shows displaced workers from AI-exposed occupations increasingly capturing large retraining returns by sorting towards less-exposed work (Hyman et al., 2025). Together these findings motivate varying both incidence and magnitude; my grid therefore extends down to small shocks.

2.3 From market incomes to disposable incomes

Whichever school proves right, the welfare consequences run through the tax-benefit system, and market-income incidence is a poor guide to disposable-income incidence. The European evidence on earlier automation makes the point directly: tax-benefit systems absorbed much of the routine-biased shock before it reached household inequality (Doorley et al., 2023). For AI, the factor-income channel compounds the case for full microsimulation: task-based models imply that automation lowers the labour share and channels income to capital owners (Acemoglu and Restrepo, 2018), Moll et al. (2022) show that automation raises returns to wealth itself, and Korinek and Stiglitz (2019) develop the theory of worker-replacing progress and the redistribution it calls for—with progressive capital taxation examined as a response by Bastani and Waldenström (2024). Because capital income is far more concentrated than labour income, this channel would tend to raise household inequality, holding the other simulated channels fixed, and only a model that jointly handles earnings, benefits, direct taxes and capital income can net the channels out.

2.4 Positioning

Exposure-based tax-benefit simulations exist, while the other incidence views are represented mainly by theory and early empirical estimates. Relative to the exposure-versus-uniform comparison in Doorley et al. (2026), I add junior-concentrated, expertise-compression and Klein-anchored stress-test allocations, hold aggregate displacement fixed across them, and examine adjustment-margin sensitivities. I report how these specified alternatives change simulated poverty, inequality and the public finances.

3 Data and methods

The analysis proceeds in three stages. I first attach an occupation-level measure of exposure to artificial intelligence to workers in a nationally grossed UK household survey. I then translate an assumed economy-wide AI shock into person-level changes to employment, wages and capital income, under alternative assumptions about who bears the displacement. Finally, I pass baseline and shocked data through an open-source tax-benefit microsimulation model to recover the distributional and fiscal consequences. This section describes each stage; the incidence scenarios that form the paper's central axis are formalised in Section 3.4.

3.1 Measuring AI exposure

My starting point is the AI Occupational Exposure (AIOE) index of Felten et al. (2021), which links progress in ten AI application areas to the occupational ability requirements catalogued in O*NET, yielding a standardised score for each occupation that captures how much of its ability bundle overlaps with what AI systems can now do. Exposure alone does not distinguish occupations in which AI substitutes for workers from those in which it augments them. I therefore combine the AIOE with the complementarity adjustment of Pizzinelli et al. (2023), who construct an index $\theta_l \in [0, 1]$ from O*NET work-context descriptors — responsibility for others, physical proximity requirements, the consequence of error — and job-zone preparation requirements. High-θ occupations are those in which AI is likely to complement human judgement rather than displace it. The complementarity-adjusted exposure measure (C-AIOE) shields high-complementarity occupations from the raw exposure score:

$$\text{C-AIOE}_l = \text{AIOE}_l \times \big(1 - (\theta_l - \theta_{\min})\big), \tag{1}$$

where $\theta_{\min}$ is the minimum complementarity across occupational groups, so that the least complementary group retains its full exposure score. Alternative exposure measures include Tolan et al. (2021), the task-level rubric of Eloundou et al. (2024), and the cross-country applications of Cazzaniga et al. (2024); Section 4 reports the sensitivity of my headline results to substituting two of these for the C-AIOE.

Constructing the index for the UK requires a bespoke crosswalk, whose provenance I state in full because no published UK C-AIOE series exists at the level I need. AIOE scores are taken from the mapping to UK SOC2020 published with the DSIT analysis of AI and UK jobs (DSIT and DfE, 2023), itself derived from the Felten et al. (2021) scores via a chained crosswalk (US SOC2018 to US SOC2010, to ISCO-08, to UK SOC2020). The complementarity indices θ are not published at occupation level, so I reconstruct them from the public O*NET 27.3 release following the recipe documented by Pizzinelli et al. (2023); the reconstruction reproduces the published cross-occupation ordering up to a level offset, which is immaterial because equations (1) and (3) use θ only relative to its minimum and its mean. Both series are aggregated to the nine SOC2020 major groups using employment weights from the Annual Survey of Hours and Earnings (ASHE) 2025 Table 14, which covers 340 of the 412 four-digit unit groups; the remainder carry no ASHE employment estimate and are excluded from the weighting. Finally, the C-AIOE scores are shifted so that the least-exposed major group scores exactly zero. Under the allocation rule in Section 3.3 that group receives no job losses, and the shift prevents negative values of the standardised index from corrupting the allocation weights.

The one-digit level of aggregation is a genuine constraint. The Family Resources Survey (FRS) person records I use carry only the one-digit SOC2020 code, so exposure varies across just nine major groups and all within-group variation is lost. The exposure gradient I impose on the population is correspondingly coarse, and results should be read with that in mind; imputation of finer occupation codes from the Labour Force Survey is a planned refinement.

3.2 Shock scenarios

The size of the aggregate shock cannot be estimated from historical data and must be assumed; I treat it as a scenario input and vary it systematically. The central calibration follows the Irish scenario design of Doorley et al. (2026): a displacement rate of 7 per cent of employees and an aggregate wage uplift of 2.6 per cent for those who remain in work. Doorley et al. convert quantities discussed by Briggs and Kodnani (2023) into these inputs; they are scenario conventions rather than estimates reported by Briggs and Kodnani or forecasts for the UK. The underlying Goldman Sachs discussion moreover concerns displacement over a roughly decade-long adoption transition with reallocation; applying the full 7 per cent within a single simulation year compresses that flow into a stock and should be read as an impact-year stress test rather than an annual forecast. Conceptually, the scenarios calibrate shocks to model primitives—occupation-level automation and productivity, as in the task framework of Acemoglu and Restrepo (2022)—whose labour-market consequences the displacement and wage rates then operationalise in reduced form. Alongside the labour-market effects, I follow Doorley et al. (2026) in raising the return to capital by 0.4 percentage points, from 1.005 per cent to 1.405 per cent, an increase they motivate using Cazzaniga et al. (2024). Moll et al. (2022) supplies only the qualitative mechanism by which automation can raise returns to wealth, not the numerical calibration used here.

Because displacement and wage growth are genuinely uncertain, the central calibration sits inside a grid spanning displacement rates from 0 to 10 per cent and aggregate wage growth from 0 to 5 per cent — 66 scenario cells in all, with the capital-return shock applied throughout. The 1 per cent "low" anchor is a pure sensitivity case: Acemoglu (2025) estimates a ten-year GDP effect of around 1 per cent, not an employment displacement rate, and I do not attribute the anchor to him. Beyond the top of the grid, the 13 per cent "high" preset stress-tests a displacement rate roughly double the central convention.

3.3 Implementing the shock at the micro level

Employment shock. The aggregate number of displaced workers is the scenario displacement rate multiplied by weighted employee employment. Following Doorley et al. (2026), the total is allocated across occupational groups l in proportion to each group's employment multiplied by its exposure. The universe is all employees: workers whose SOC code cannot be matched to an exposure score form a pseudo-group assigned the employment-weighted mean exposure, so no employee is mechanically excluded from risk:

$$\text{JobLoss}_l = \text{TotalJobLoss} \times \frac{\text{EMP}_l \times \text{C-AIOE}_l}{\sum_l \text{EMP}_l \times \text{C-AIOE}_l}, \tag{2}$$

so job losses concentrate in large, highly exposed groups and the zero-scored least-exposed group loses none. Within each group, displaced individuals are selected by systematic sampling on a random permutation of the group's members: the weighted quota from equation (2) fixes each record's first-order inclusion probability, which is equal for all records within a group (up to the prescribed probability tilts, such as the youth multiplier, which enter as tilts to these inclusion probabilities), and the systematic pass realises the quota exactly in expectation. A record's inclusion probability therefore does not depend on its grossing weight; the weights enter only through the quota accounting, which is the design intent.

Displacement is modelled as full withdrawal from employee work. A displaced worker's employment income, hours, employee pension contributions and salary-sacrifice amounts are set to zero, any statutory pay is removed, and labour-market status is set to unemployed before the shocked simulation runs. Dual jobholders retain self-employment income. This differs from Doorley et al. (2026), who use a labour-market adjustment add-on to impute unemployment duration and work history for transition cases. The full-year zero-earnings treatment here is an author-chosen UK calibration; the duration sensitivity is a 50 per cent earnings-retention hybrid that leaves status and hours at full-year unemployment, not a modelled six-month spell. Task displacement can instead manifest as wage change and reallocation (Acemoglu and Restrepo, 2022), and the wage-margin family examines that possibility. These alternatives do not bound all adjustment paths. PolicyEngine UK takes no unemployment-duration input, so displaced workers are assessed as current-period unemployed under standard benefit rules.

Wage shock. Each worker who survives the employment shock receives a percentage uplift proportional to complementarity, normalised by the employment-weighted mean complementarity over baseline workers:

$$\text{WageChange}_l = \text{WageGrowth} \times \frac{\theta_l}{\bar{\theta}} \times \text{Wage}_l, \qquad \bar{\theta} = \frac{\sum_l \text{EMP}_l\, \theta_l}{\sum_l \text{EMP}_l}, \tag{3}$$

where EMP_l is baseline (pre-shock) employment. This is the baseline wage rule. Two properties should be noted. The normalisation makes the *employment-weighted mean* percentage uplift equal the assumed wage growth. The aggregate wage bill grows by exactly that rate only if θ is uncorrelated with pay. Since high-θ occupations earn more, the wage-bill-weighted uplift is somewhat larger. A second property: because $\bar{\theta}$ is computed over all baseline workers while the uplift accrues only to survivors—who, being drawn from less-exposed occupations, have above-average θ—displacement raises the realised mean survivor uplift slightly above the headline rate. Exact conservation is verified in seeded tests at zero displacement; both departures are properties of the rule itself rather than implementation error. The economic logic mirrors equation (1): AI complements rather than substitutes for high-θ occupations, so those occupations capture a larger share of the productivity gains — an assumption the compression scenario of Section 3.4 deliberately inverts. In the decomposition of Acemoglu and Restrepo (2022), this uplift plays the role of the economy-wide productivity effect, while the displacement effect is carried here entirely by the employment margin; their ripple effects — displaced workers bidding down the wages of the non-displaced — are outside the baseline framework, so non-displaced workers can only gain; Appendix A.8 prices this channel in a reduced-form sensitivity.

Capital shock. The 0.4 percentage-point increase in the return to capital is implemented by scaling each person's savings-interest and dividend income by the ratio of shocked to baseline return, $1.405/1.005 \approx 1.398$. The baseline return of 1.005 per cent is taken from the baseline calibration; because the scaling factor is the ratio of two return levels, the size of the capital channel is highly sensitive to this assumption. The same +0.4 percentage-point shock applied to, say, a 4 per cent baseline return would scale capital incomes by only 10 per cent rather than 40 per cent. The capital-channel results should therefore be read as conditional on this calibration; equivalently, the shock can be read directly as a 39.8 per cent proportional increase in interest and dividend income. In the framework of Moll et al. (2022) the relevant return is that on investors' wealth rather than the safe rate, so a refinement would load the shock onto dividend income alone; loading the shock on dividends alone could produce a more top-concentrated incidence, but I do not quantify that alternative. Rental income is excluded from the scaling, on the modelling assumption that the AI productivity channel operates through financial rather than housing capital. The self-employed are outside both labour-market shocks; their interest and dividend income is scaled like everyone else's.

3.4 Incidence scenarios

The rules above fix how large the shock is; they do not settle who bears it, and the empirical literature has not settled that question either. I therefore treat incidence as an explicit scenario dimension. Holding the central aggregate shock fixed—7 per cent displacement, a 2.6 per cent wage uplift, the capital shock—I re-allocate the same weighted quota of displacement under four stylised incidence families, plus a fifth Klein-anchored stress test introduced below:

Exposure-proportional. The default allocation of equations (2) and (3): displacement follows the C-AIOE gradient and wage gains follow complementarity. This family is the central calibration used throughout the shock-size grid.

Junior-concentrated. Displacement draws are tilted towards workers aged 16–24 by a multiplier equal to the ratio of junior to overall displacement effects reported by Klein Teeselink (2025) for the UK (5.8/4.5), consistent with the seniority-biased evidence of Hosseini Maasoum and Lichtinger (2026a) and Brynjolfsson et al. (2025a). The group quotas and wage rule are otherwise unchanged.

Expertise-compression. A stylised, author-designed stress test of the view that AI compresses expert rents rather than displacing routine work; as implemented it is a *high-earner displacement tilt*, and the label "expertise-compression" names the motivating hypothesis rather than a modelled compression mechanism. Displacement probabilities are tilted (multiplier 2, an author choice) towards the top weighted-earnings tertile of workers in above-median-exposure occupations. The survivor-wage and capital channels remain exactly those of the exposure-proportional family, so the headline factorial comparison changes only who is displaced. An older compound variant that also reflects θ about its range is retained in the replication outputs but is not part of the headline comparison. The formal treatment in Hosseini Maasoum and Lichtinger (2026b) operates through wage compression for continuing workers rather than job destruction; this displacement-based implementation is therefore a fiscal stress test, not an implementation of their model.

Uniform. Every employee has the same first-order displacement probability, while the central complementarity-graded survivor-wage channel is held fixed. Relative to exposure-proportional incidence, it therefore isolates a flat displacement allocation. An earlier bundle that flattened both displacement and wages is retained in the replication outputs but is not reported here.

Each displacement family has the same expected weighted count of displaced workers. Cross-family differences in the headline factorial comparison reflect displacement incidence alone; survivor wages and capital income are held fixed at the person level. These rules are scenario assumptions, not estimates of realised AI displacement.

Klein-anchored stress test. Section 4.5 adds a fifth, explicitly author-constructed family loosely anchored to the UK firm-level evidence of Klein Teeselink (2025). Displacement probability is proportional to shifted C-AIOE multiplied by a junior factor of 1.29 and wage-tier factors of 9.6 above versus 0.6 below weighted-median person earnings; the aggregate shock stays at the central preset. These values are not a transported estimate of household-level displacement. The December 2025 revision of Klein Teeselink (2025) reports per-standard-deviation semi-elasticities of about -0.3 per cent overall, -0.4 per cent for junior employment, and -0.7 per cent in high-compensation firms against a statistical zero in low-compensation firms. The much larger wage-tier factors used here descend from press-release scalings to maximally exposed firms, and person earnings only proxy the paper's firm-average compensation split. No geographic multiplier is applied: the same revision reports minimal, statistically insignificant London effects. The remaining junior proxy (age under 25), wage-tier mapping and multiplicative combination are author choices, so this family is a top-loaded stress test, not a measured wage or displacement schedule.

Wage margin. The families above vary who is displaced; a final family varies the *margin* on which the shock arrives, and I treat it as a separate adjustment-margin robustness family rather than a sixth incidence family, since it changes the mechanism rather than the allocation. In the task framework of Acemoglu and Restrepo (2022), displacement "does not need to be associated with job loss" and manifests chiefly as relative wage declines for workers who stay employed; the entry-barrier mechanism of Hosseini Maasoum and Lichtinger (2026b) likewise operates through wage adjustment rather than job destruction. The wage-margin family moves the same aggregate earnings loss to that channel: no worker loses a job, and instead every employee's earnings are cut by an occupation-level percentage proportional to $\max(0, \text{exposure} - \text{exposure}_{\min})$ — the same min-shift normalisation as the quota weights of equation (2), so the least-exposed group takes no cut—with a single calibration constant chosen so that the weighted aggregate employee-earnings loss equals the *realised* gross earnings loss of the paired central displacement draw for the same seed (a realised paired share of 7.77 per cent of the baseline employee wage bill at seed 0, larger than the 7 per cent head-count rate because group job-loss rates correlate with group pay). The family therefore shares its calibration draw-by-draw with the central scenario rather than with a fixed 7 per cent wage-bill share, and its only seed variation enters through the paired draw's realised loss. I run two gradients: the C-AIOE itself, and the potential-supply-shift (PSS) measure of Hosseini Maasoum and Lichtinger (2026b), taken from their public data release and crosswalked to SOC2020 major groups by the same chained SOC2018–SOC2010–ISCO-08–SOC2020 route with ASHE 2025 employment weights as the exposure scores (Spearman rank correlation with the C-AIOE across the nine major groups: 0.78). The PSS rows should be read as a *PSS-ranked wage-cut stress test*: PSS supplies only the cross-occupation ranking of author-imposed cuts, and the exercise is not an implementation of the Hosseini Maasoum and Lichtinger (2026b) labour-supply mechanism, which their citation here motivates rather than licenses. The 2.6 per cent complementarity-graded wage uplift of equation (3) and the capital shock compose on top, so a worker's net percentage change is uplift minus cut; employment status, hours and pension contributions are unchanged. Section 4.6 reports the results.

Mixed adjustment and adverse wage pass-through. Two further sensitivities relax the assumption that adjustment occurs entirely on one margin and that workers who retain their jobs necessarily gain. First, for each seed I calculate the gross pre-uplift earnings removed by the central 7 per cent employee-displacement draw and hold that amount fixed. Let $\lambda \in \{0, .25, .5, .75, 1\}$ scale the employee-displacement rate, 0.07λ. C-AIOE-graded cuts among survivors then fill the exact gap between earnings removed through partial displacement and the seed-specific central gross-loss target. The usual 2.6 per cent complementarity-graded uplift and capital shock compose only after this identity is imposed. Thus $\lambda = 1$ exactly reproduces the central displacement scenario for the same seed; $\lambda = 0$ is a pure wage-cut counterfactual matched to the central draw's realised loss, not the separate family that removes exactly 7 per cent of the baseline wage bill. The intermediate cells are author-designed reduced-form comparative statics rather than equilibrium transition paths. Second, holding the central 7 per cent displacement draw fixed, I replace the assumed average survivor-wage effect with $\{-5, -2.6, 0, +2.6, +5\}$ per cent, retaining the complementarity gradient. This grid includes wage suppression, no pass-through, the central calibration and a larger productivity pass-through without requiring a model of firm wage-setting or bargaining.

3.5 Microsimulation with PolicyEngine UK

I use PolicyEngine UK (version 2.89.2), an open-source tax-benefit microsimulation model, running on PolicyEngine UK's processed (base, unenhanced) Family Resources Survey 2024–25 microdataset. This reshapes the UK Data Service FRS 2024–25 (study SN 9563) into PolicyEngine's tax-benefit schema, applies source-level imputations (council tax, benefit take-up, salary sacrifice) and retains the survey's own grossing weights; I obtain it from PolicyEngine's public data repository. Occupation codes are not carried in the processed dataset, so I join the SOC2020 major group from the raw FRS adult file (UK Data Service SN 9563) onto the simulation's person table using the survey's household and person identifiers, which match every FRS adult. The simulation year is 2026.

The counterfactual design is a paired comparison on identical data: a baseline simulation and a shocked simulation are run on the same records, with the shocked run receiving the modified employment income, hours, pension contributions, savings-interest income, dividend income and employment status of Section 3.3 as direct inputs. All reported effects are differences between the two runs. This design removes between-run noise — the survey sample and grossing weights are identical on both sides, so no part of a reported effect reflects resampling — but it does not quantify population sampling uncertainty: both runs inherit whatever sampling error the FRS itself carries, and my Monte Carlo intervals speak only to the stochastic identity of the displaced, not to survey sampling.

All income results use the HBAI cash disposable income concept. Absolute BHC and AHC poverty compare equivalised household income with fixed real thresholds derived from the baseline; those thresholds are frozen in baseline–shock comparisons. Relative BHC and AHC poverty instead use 60 per cent of the contemporaneous median equivalised income, so their thresholds move with the simulated distribution. I report the government-balance change; person-weighted BHC and AHC poverty; the Gini of equivalised household disposable income, with negative incomes bottom-coded at zero; and mean household-income changes by baseline-fixed decile.

Because the identity of the displaced is stochastic, I adopt explicit seed conventions and label every result accordingly. The 66-cell grid, the three presets and the decomposition figure are single draws at seed 0, so that cells differ only in their scenario inputs, never in their random draw. The five incidence families and policy comparisons report means over 50 paired draws (seeds 0–49), with BHC poverty, AHC poverty and the Gini evaluated on every draw. Their machine-readable summaries report means, assignment-draw standard deviations and quantiles separately from Monte Carlo standard errors, plus convergence comparisons against the original 20-draw budget. The decile transition and wage profiles (Figures 1 and 2) also average 50 seeded draws. The duration, take-up and decomposition sensitivities retain separately declared 20-draw budgets; wherever a ± value appears it is the standard deviation across draws, not a survey-based standard error.

Policy reforms in the shocked world. The reforms of Section 5 are evaluated as $M(\text{shock}+\text{reform}) - M(\text{shock})$ on the fixed seed-0 shock draw, applying for 2026 only, with poverty lines held at their shocked-world values. Benefit-side reforms are implemented as PolicyEngine UK parameter reforms — scaling the Universal Credit (UC) standard allowance, raising the benefit-cap thresholds and cutting the taper rate — so their costs are net of the means-testing clawback the model computes. Wage insurance is instead a direct post-simulation transfer to displaced workers (half of lost earnings, capped), treated as non-taxable and disregarded by means tests, so its gross cost equals its net cost and brackets the generous end of possible implementations.

4 Results

I first report the central scenario and its household mediation, then vary shock size, incidence and adjustment margin. Seed and draw conventions follow Section 3.5; ± values are standard deviations across assignment draws, not sampling standard errors.

4.1 The three market-income channels

The central scenario displaces 7 per cent of employees—1.56 million workers, allocated across occupations in proportion to exposure—and grants surviving workers in exposed occupations a wage uplift averaging 2.6 per cent, alongside the capital income channel. Figure 1 plots the share of each equivalised disposable income decile that transitions from employment to unemployment. The gradient is strictly monotone: 0.50 per cent of the bottom decile transitions, rising through every decile to 4.89 per cent at the top. It is important to be clear about what this figure shows. The gradient is not an empirical finding about who AI will in fact displace; it is the exposure-proportional allocation assumption made visible. Because the professional, associate-professional and administrative occupations that score highest on the C-AIOE index sit in the upper half of the income distribution, allocating displacement in proportion to occupational exposure necessarily concentrates job loss among higher-income households. Doorley et al. (2026) obtain a similarly top-concentrated simulated profile for Ireland using different microdata and a finer occupational classification. Both results are conditional on exposure-proportional allocation and should not be read as evidence of realised incidence. Section 4.5 therefore varies that assumption.

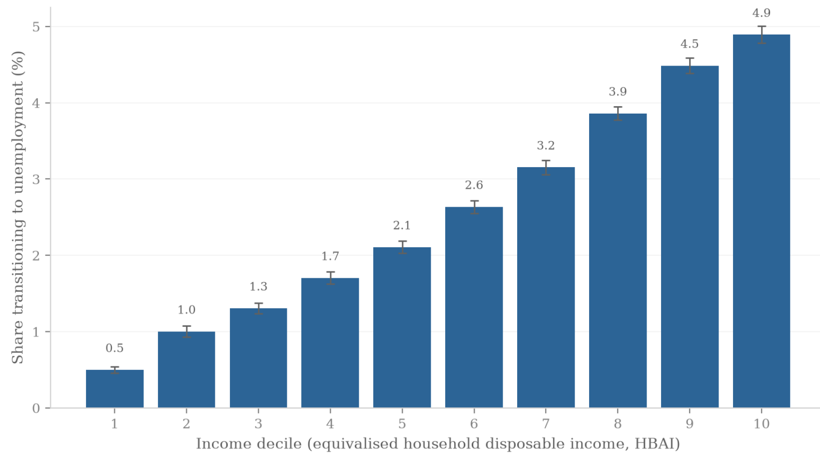

Figure 1: Share of the population transitioning from employment to unemployment by equivalised disposable income decile, central scenario. Bars are means over 50 seeded displacement draws; error bars show central 95 per cent intervals across assignment draws. The gradient reflects the exposure-proportional allocation assumption, not observed incidence (see text).

The wage channel, by contrast, is almost distributionally flat. Figure 2 shows the percentage change in employment income among workers who remain employed: gains run from 2.50 per cent in the bottom decile to 2.72 per cent in the top, a spread of barely 0.2 percentage points around the 2.6 per cent average uplift. Exposure is sufficiently widespread across the deciles in which people work that the productivity dividend is spread thinly and evenly. Under the central design, most cross-decile variation comes from the displacement margin. That allocation is a property of the scenario design rather than a finding: in the US evidence of Acemoglu and Restrepo (2022), task displacement operates chiefly through relative wage declines, which account for 50–70 per cent of changes in the wage structure. Section 4.6 examines this alternative: a wage-margin implementation of the same headline calibration shifts losses from benefit caseloads to earnings, with a comparable fiscal cost but an almost entirely different poverty and inequality profile.

The third channel is capital income. The scenario assumes part of the productivity gain is capitalised into returns on existing capital, applying a uniform proportional uplift to interest and dividend income (Section 3.3). Although the percentage change is identical across deciles by construction, Figure 3 shows the incidence in pounds is heavily concentrated: the top decile holds 35.2 per cent of all capital income (£8.1 billion of the baseline aggregate in my data), the top two deciles together more than half, and the bottom decile just 2.8 per cent. Under this proportional allocation, most simulated capital-income gains accrue to households already near the top of the distribution.

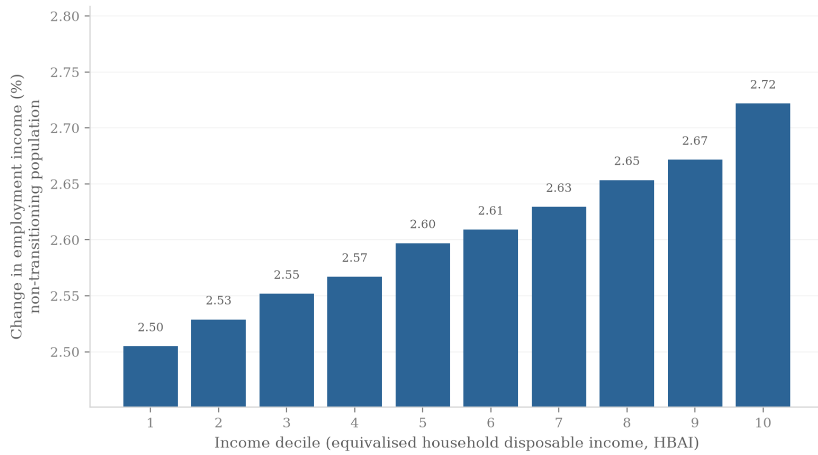

Figure 2: Percentage change in employment income among workers remaining in employment, by income decile, central scenario. Means over 50 seeded draws.

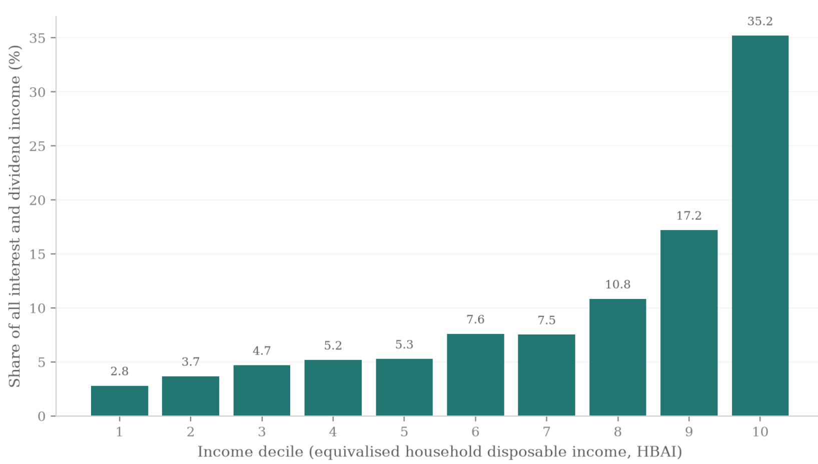

Figure 3: Capital income channel by decile: a uniform proportional uplift, but pound gains concentrated where capital income is held.

4.2 Household mediation of the shock

The tax-benefit system converts each market-income loss into a smaller disposable-income loss; Figure 4 decomposes that mediation by decile into market income, benefits, and taxes and contributions, all expressed relative to baseline disposable income. The decomposition is household-level and additive: market income change plus benefit change minus tax change equals the disposable income change up to an explicit residual, which collects perimeter items inside the HBAI concept (council tax and pension deductions) and never exceeds 0.5 per cent of baseline income in any decile.

Market income losses follow the transition gradient: they are near zero in the bottom decile, around 0.8–2.0 per cent of baseline disposable income in deciles 3–5, between 3.3 and 4.1 per cent in deciles 7–9, and reach 6.7 per cent in the top decile. The tax-benefit system then cushions these losses through two distinct mechanisms. Across the middle deciles, automatic benefit entitlements—Universal Credit in particular—rise by up to 0.4 per cent of baseline income as displaced earners qualify for means-tested support. At the top the cushioning operates through the tax system instead: deciles 7–10 see their income tax and National Insurance liabilities fall by 0.4–2.0 per cent of baseline income as high-marginal-rate earnings disappear. The combined offset is substantial. In decile 10, a market income loss of 6.7 per cent of baseline disposable income becomes a disposable income loss of 4.3 per cent—roughly a third of the market shock absorbed—and in decile 5 a 1.8 per cent market loss becomes a 1.1 per cent disposable loss. Averaged over 20 displacement draws, disposable income losses run from 0.75 per cent in the bottom decile to 3.87 per cent in the top; the gradient is upward and stable across draws, though on the current 20-draw means it is not everywhere strictly monotone: decile 3 loses slightly less than decile 2 (-0.85 against -0.93 per cent) and deciles 9 and 10 are essentially tied (-3.87 per cent each), both differences well inside the draw noise.

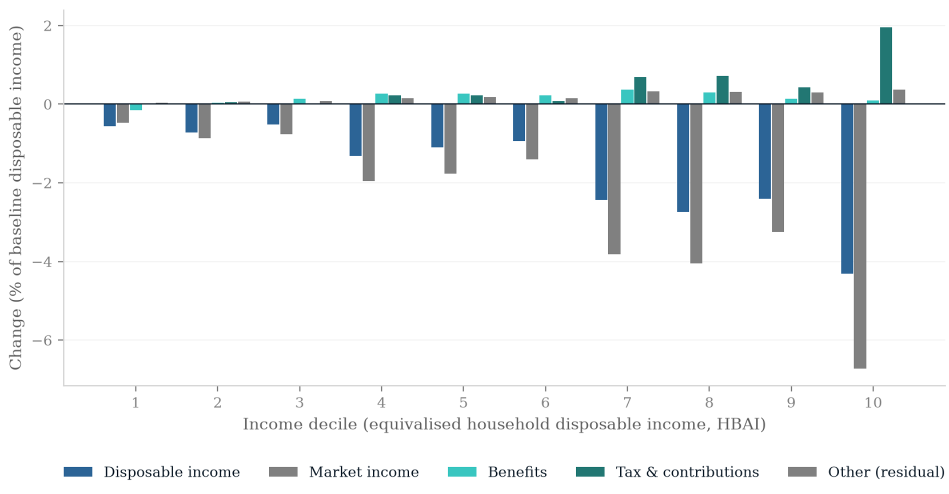

Figure 4: Additive household-level decomposition of the change in income by decile into market income, benefits, and taxes and contributions, central scenario (single draw, seed 0), as a share of baseline HBAI disposable income. The residual collects perimeter items (council tax, pension deductions) inside the HBAI concept.

4.3 Fiscal and distributional aggregates

Table 1 summarises the preset scenarios. In the central scenario the annual Exchequer cost is £18.2 billion, driven by lost income tax and National Insurance on displaced higher earners together with higher benefit spending. Poverty rises by 1.81 percentage points before housing costs (1.85 points after housing costs) and the Gini coefficient of equivalised disposable income rises by 1.04 percentage points from a baseline of 0.309. The low scenario—a 1 per cent displacement rate with no wage gain, which I treat purely as a sensitivity case rather than an estimate drawn from the literature—costs £4.3 billion and moves poverty and the Gini by roughly 0.2–0.3 points each. The high scenario, at 13 per cent displacement, costs £46.9 billion a year with poverty up 4.05 points. PolicyEngine's before-housing-costs poverty line is an *absolute* threshold, so these poverty changes measure absolute impoverishment rather than movement relative to a shock-shifted median. On the relative (60 per cent of contemporaneous median) BHC line the central scenario raises poverty by $+0.71 \pm 0.16$ percentage points over 50 draws — well under half the absolute-line rise, because the shock lowers the median itself and so drags the relative line down with it.

The scenario anchors are calibration conventions rather than forecasts (Section 3.2); in particular, the low scenario is not an employment estimate from Acemoglu (2025), whose 0.9–1.1 per cent figure refers to ten-year GDP growth.

The central poverty response is large by the standard of recent UK downturns, in which absolute poverty barely moved, so it deserves decomposition. The +1.81-point net change is a gross inflow of 1.32 million people (+2.10 points), entirely from households containing a displaced worker, net of a 0.18-million outflow (−0.29 points) lifted over the line by the capital and survivor-wage channels. The entrants are not marginal cases nudged across a threshold: they come from across baseline deciles 2–10 (11 per cent from the top decile, where a displaced sole earner falls the full distance), their households start at a median 1.81 times the poverty line, and they land at a median 41 per cent below it. Sixty-one per cent depended entirely on the displaced worker's earnings. The magnitude therefore follows from the framework's polar assumptions—full-year earnings loss and low Universal Credit replacement rates against a baseline-fixed line—rather than from mass bunched just above the threshold; the duration and wage-margin sensitivities below show how quickly it attenuates when those assumptions are relaxed, and a 2008–09-sized backcast in Appendix A.9 quantifies how much of a recession's measured poverty response the excluded uprating and duration channels absorb in practice.

Table 1: Summary of preset scenarios (single draw, seed 0)

Scenario	Displaced (million)	Exchequer (£bn/yr)	Δ Poverty (BHC, pp)	Δ Gini (pp)
Low	0.24	4.3	+0.33	+0.21
Central	1.56	18.2	+1.81	+1.04
High	3.04	46.9	+4.05	+2.07

Note: Low: 1 per cent displacement, no wage gain (sensitivity case). Central: 7 per cent displacement, 2.6 per cent wage gain. High: 13 per cent displacement, 2.6 per cent wage gain. The capital-return shock (+0.4pp) applies in every preset.

The central estimates are similar, but not identical, across the 50 assignment draws: the mean Exchequer cost is £17.7 billion (assignment-draw standard deviation £1.2 billion; range £15.2–20.3 billion), with mean poverty and Gini increases of 1.87 ± 0.15 and 1.11 ± 0.09 percentage points. In the alternative indices tested, headline estimates change moderately: replacing C-AIOE with the DSIT AIOE mapping leaves the seed-0 Exchequer cost essentially unchanged, while the GPT-exposure measure of Eloundou et al. (2024) raises it by about £2.2 billion (12 per cent); both produce similar upward transition gradients (Appendix A.7).

Two tested assumptions materially change the headline numbers. The first is displacement duration. The central scenario removes a full year of earnings from every displaced worker; under a 50 per cent earnings-retention hybrid—annual earnings are halved while employment status and hours remain those of full-year unemployment, so this approximates rather than models a six-month spell—the Exchequer cost falls from £17.7 billion to £4.0 $\pm$ 0.7 billion and the BHC poverty rise all but vanishes ($+0.17 \pm 0.10$ points), over 20 draws. The collapse is more than proportional—a naïve halving of the full-year result would give £8.9 billion—because workers who keep half a year's earnings remain taxpayers and largely stay clear of means-tested support. Every fiscal and poverty magnitude in this paper therefore scales strongly with the expected out-of-work duration, and the headline figures should be read as a full-year no-reemployment case. The second is benefit take-up. Imposing a 70 per cent Universal Credit take-up probability (drawn identically in baseline and shocked worlds) moves the cost to £18.5 $\pm$ 1.4 billion and the poverty rise to $+1.76 \pm 0.13$ points: the headline estimates change little under this single take-up sensitivity.

4.4 The shock-size grid

To move beyond the presets I simulate a grid of 66 scenarios, crossing eleven displacement rates (0 to 10 per cent of employees, allocated in proportion to exposure, as in the presets) with six wage uplifts (0 to 5 per cent), each combined with the capital income channel, all under exposure-proportional incidence. Figure 5 maps the change in average household disposable income. The corners bracket the range: with a 5 per cent wage gain and no job loss, average disposable income rises by 3.0 per cent; with 10 per cent displacement and no wage gain it falls by 5.1 per cent. The near-zero contour runs close to the grid diagonal—at these calibrations, roughly one percentage point of wage growth offsets one percentage point of displacement in aggregate disposable income terms—so whether AI raises or lowers average living standards is highly sensitive, within this grid, to the balance struck between these two forces, about which the empirical literature remains genuinely uncertain.

Figure 6 shows the corresponding fiscal picture. One measurement issue deserves note: in PolicyEngine's UK aggregates, income tax plus National Insurance minus total benefit expenditure (which includes the state pension) is a negative number, so expressing revenue changes relative to that net position produces uninformative percentages. I therefore express the net revenue change relative to gross income tax and National Insurance receipts as simulated on the household microdata (about £313 billion); this base is smaller than headline OBR receipts, which include employer contributions and income outside the household survey universe. Two consequences follow. First, the grid's revenue measure is income tax plus National Insurance minus cash benefit expenditure, a narrower concept than the full government-balance measure behind the £18.2 billion headline cost of Section 4.3, which also includes indirect taxes and other levies; the grid's cash-tax cost of the central 7 per cent/2.6 per cent scenario is roughly £11 billion (interpolated between the 2 and 3 per cent wage columns), and the two figures should not be read off the same surface. Second, on that narrower basis the grid spans a gain of £22.3 billion, or 7.2 per cent of receipts (5 per cent wage growth, no displacement), to a loss of £29.4 billion, or 9.4 per cent of receipts, in the worst cell (10 per cent displacement, no wage growth).

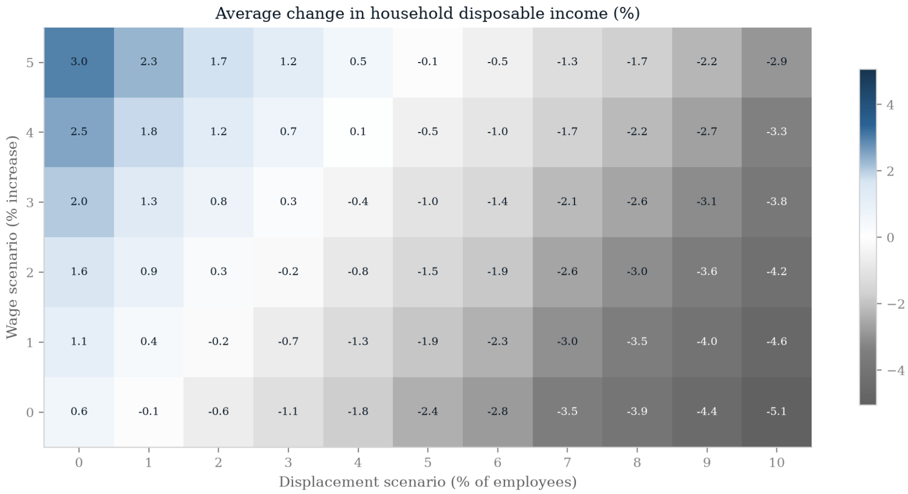

Figure 5: Change in average household disposable income (per cent) across the 66-cell grid of displacement rates and wage uplifts (single draw, seed 0).

At the central displacement rate, 7 per cent with no wage offset costs £21.2 billion (6.8 per cent of receipts), which a 3 per cent wage uplift more than halves to £9.8 billion. As with disposable income, the break-even locus lies close to the diagonal.

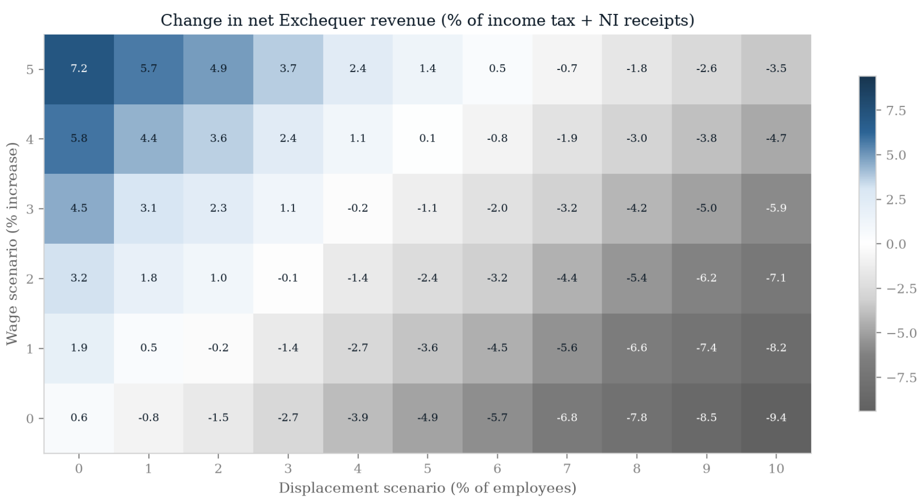

Figure 6: Net Exchequer revenue change as a percentage of gross income tax and National Insurance receipts, across the 66-cell scenario grid.

Figure 7 maps the change in the Gini coefficient across the same grid, and produces a consistent result within this grid: the Gini rises in all 66 cells, from +0.06 percentage points in the capital-only cell (no displacement, no wage gain) to +1.61 points at the far corner. No combination of parameters within the grid reduces measured inequality. The simulated pattern is consistent with widening gaps at both ends of the distribution rather than a simple top-income gain, and it coexists with an incidence that is itself progressive: displacement strikes hardest at the top of the earnings ladder and pushes affected households down the distribution, while the wage uplift lifts the surviving employed—disproportionately in the upper-middle and top deciles—further away from workless and low-income households, and the capital channel reinforces the very top. Within the model, losses and gains are allocated in ways that widen gaps between households, and inequality rises even though the market shock lands mainly on higher earners.

Figure 7: Change in the Gini coefficient of equivalised HBAI disposable income (percentage points) across the 66-cell scenario grid. The Gini rises in every cell.

4.5 The incidence axis: who bears the shock

Everything so far conditions on exposure-proportional incidence. But allocation is at least as uncertain as shock size, so I reallocate the same central calibration across five families. Four are stylised: exposure-proportional, junior-concentrated, expertise-compression (implemented as a high-earner displacement tilt, not the Hosseini Maasoum and Lichtinger (2026b) wage mechanism; Section 3.4) and uniform. The fifth is a Klein-anchored top-loaded stress test: exposure-proportional draws multiplied by an author-imposed junior tilt (1.29) and wage-tier tilt (9.6 above, 0.6 below median person earnings). Table 2 gives means and assignment-draw standard deviations over 50 paired seeds.

Within these calibrations, who bears the shock materially changes its simulated cost. The mean Exchequer cost ranges from £13.9 billion under uniform incidence to £32.8 billion under the Klein-anchored stress test, generated by reallocating which workers lose their jobs. The larger fiscal effects under top-loaded incidence are consistent with the larger tax liabilities of the displaced high earners. These are paired-seed scenario comparisons, not population confidence statements.

Across the four stylised families, the mean BHC poverty rise lies between 1.85 and 1.96 percentage points, while mean Exchequer costs span nearly £7 billion. Because the families share assignment seeds, draw-level *differences* support paired inference: the expertise-compression poverty effect exceeds the exposure-proportional one by +0.09 points (paired 95 per cent interval [+0.03, +0.15]). The uniform–exposure poverty difference is indistinguishable from zero

Table 2: Central shock under five incidence calibrations

Incidence family	Displaced (million)	Exchequer cost (£bn/yr)	Δ Poverty BHC (pp)	Δ Gini (pp)
Exposure-proportional	1.56	17.7 ± 1.2	+1.87 ± 0.15	+1.11 ± 0.09
Junior-concentrated	1.55	17.9 ± 1.6	+1.85 ± 0.14	+1.08 ± 0.08
Expertise-compression	1.67	20.6 ± 1.8	+1.96 ± 0.15	+1.04 ± 0.09
Uniform	1.55	13.9 ± 1.7	+1.87 ± 0.17	+1.16 ± 0.10
Klein-anchored stress test	1.62	32.8 ± 2.2	+2.24 ± 0.14	+0.97 ± 0.12

Note: Means ± assignment-draw standard deviations over 50 paired seeds. Assignment quantiles and Monte Carlo standard errors are reported separately in the machine-readable summary; they describe allocation noise, not population uncertainty. The Klein-anchored family combines exposure with author-imposed junior (1.29) and wage-tier (9.6 above / 0.6 below median person earnings) multipliers. The wage-tier values use press-release maximum-exposure scalings rather than the paper's per-standard-deviation estimates; this is a stress test, not an estimated displacement schedule. AHC poverty is evaluated on the same 50 draws as BHC poverty and the Gini.

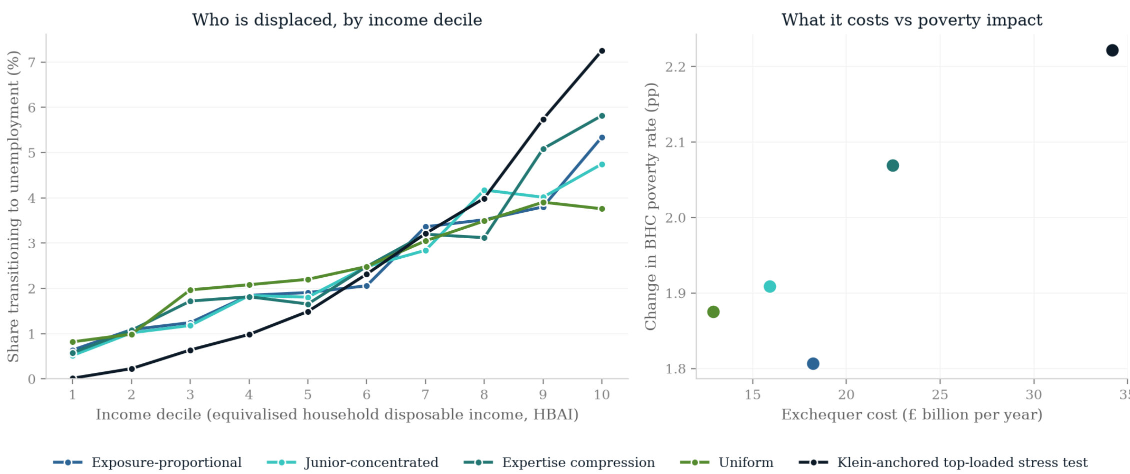

Figure 8: Decile incidence of the same aggregate central shock under five incidence families: share of each decile transitioning to unemployment, and change in equivalised HBAI disposable income by decile. The figure shows a single illustrative draw (seed 0), whereas the aggregates in Table 2 are means over 50 Monte Carlo draws.

(-0.00 points, $[-0.06, +0.06]$); all families share the central wage and capital axis, so the comparison isolates displacement allocation. Uniform incidence retains a reliably smaller fiscal effect than exposure-proportional incidence ($-\pounds3.84$ billion, $[-4.46, -3.21]$) alongside a reliably larger Gini effect (below). The junior-concentrated and exposure-proportional fiscal costs remain indistinguishable under the same paired comparison ($+\pounds0.19$ billion, $[-\pounds0.30, +\pounds0.67]$). These paired intervals describe assignment-draw noise only, not population sampling uncertainty.

The author-imposed top-loaded calibrations combine relatively large fiscal and poverty effects. Expertise compression costs £20.6 billion and raises BHC poverty by +1.96 points. The Klein-anchored stress test has means of £32.8 billion and +2.24 points; its AHC effect is evaluated over the same 50 assignment draws. Its wage-tier multiplier produces the most top-tilted seed-0 gradient, from 0.01 per cent transitioning in the bottom decile to 7.25 per cent in the top. These magnitudes are conditional on the stress-test mapping and are not estimates transported from Klein Teeselink.

The Gini rises in all five simulated families—50-draw means of +0.97 to +1.16 percentage points—just as it rises in all 66 grid cells (in the appendix's capital-off variant of the grid the null cell with no shock at all is exactly zero, as it must be). Paired inference again separates some families: the uniform Gini effect exceeds the exposure-proportional one by +0.04 points (paired 95 per cent interval $[+0.01, +0.08]$), and the Klein-anchored Gini effect is reliably *smaller* than the expertise-compression one (-0.07 points, $[-0.12, -0.03]$), consistent with its more top-concentrated losses compressing the upper tail. The directional result is explicitly conditional on these displacement calibrations.

4.6 The adjustment margin: wage cuts versus displacement

The incidence families reallocate displacement across workers; the wage-margin family instead moves the same shock to the wage margin. Every employee keeps their job, and earnings are removed through author-imposed occupation-level cuts proportional to C-AIOE or PSS, calibrated so that the aggregate employee-earnings loss equals the realised gross earnings loss of the paired central displacement draw—7.77 per cent of the baseline wage bill at seed 0 (recorded target share 0.07774), larger than the 7 per cent head-count rate because group job-loss rates correlate with group pay; the equality is asserted in code. The PSS rows are a PSS-ranked wage-cut stress test, not the Hosseini Maasoum and Lichtinger (2026b) labour-supply mechanism. Table 3 compares these clearly labelled calibrations on the seed-0 basis.

Table 3: The headline calibration on two adjustment margins (central calibration; displacement row single draw, seed 0; wage-margin rows deterministic)

Scenario	Exchequer cost (£bn/yr)	Δ Poverty BHC (pp)	Δ Gini (pp)
Central displacement	18.2	+1.81	+1.04
Wage margin (C-AIOE gradient)	26.6	+0.10	−0.44
Wage margin (PSS gradient)	26.4	+0.17	−0.35

Note: The wage-margin rows remove the same aggregate gross earnings as the paired seed-0 central displacement draw (7.77 per cent of the baseline employee wage bill); the displacement row removes the earnings of a weighted 7 per cent of employees. All rows apply the 2.6 per cent wage uplift and the capital shock. The wage-margin scenarios displace no one: cuts are occupation-level percentages proportional to the stated gradient, calibrated to the stated baseline-wage-bill share. After-housing-costs poverty changes are +1.85, +0.27 and +0.35 points respectively.

Three contrasts stand out. First, under the simulated wage schedules the fiscal cost rises from £18.2 billion to £26.6 billion (£26.4 billion under the PSS gradient). Wage cuts shave earnings at marginal tax and National Insurance rates, whereas displacement removes whole earnings blocks and is partly offset by means-tested benefits. This comparison is conditional on these calibrations and does not establish a general lower bound.

Second, the poverty effect all but disappears in the simulated wage-cut cases: the +1.81-point BHC poverty rise becomes +0.10 points under the C-AIOE gradient and +0.17 under PSS. Spreading the calibrated loss across employees pushes fewer households across the frozen absolute line than concentrating losses on displaced workers; this does not establish a bound for other wage schedules.

Third, the sign of the inequality effect flips under both author-chosen wage gradients: the Gini falls by 0.44 and 0.35 points, whereas it rises in the simulated displacement cells. Percentage cuts scale with gradients concentrated in well-paid occupations, compressing earnings from above. The comparison shows that poverty and inequality conclusions depend on both the adjustment margin and its imposed schedule; it does not establish a universal fiscal or distributional result.

Table 4 makes the like-for-like comparison. In every row gross earnings loss before the common uplift is 7.77 per cent of the baseline wage bill, exactly the loss realised in the seed-0 central/exposure draw to which the exercise is paired. Moving that fixed loss from wage cuts to displacement reduces the Exchequer cost from £26.6 billion to £18.2 billion, while raising BHC poverty from 0.10 to 1.81 points and the Gini effect from −0.44 to +1.04 points. The half-index case costs £22.7 billion, raises poverty by 1.02 points and raises the Gini by 0.29 points. Net employee-income losses after the uplift rise slightly, from 5.11 to 5.32 per cent, because fewer survivors receive the uplift as displacement rises; the pre-uplift loss being compared is invariant. Conditional on this author-designed allocation rule, the monotone poverty and inequality patterns track the shifting adjustment margin; they are not structural causal estimates.

Allowing survivor wages to fall is more consequential for the level of the fiscal estimate. With the same 1.56 million displaced workers, removing the productivity pass-through raises the Exchequer cost from £18.2 billion to £31.6 billion; an average 2.6 per cent wage decline raises it to £45.0 billion, and a 5 per cent decline to £57.3 billion. Poverty also rises, but less dramatically, from 1.81 points under the central uplift to 2.65 points under a 5 per cent wage decline, because the additional loss is spread across survivors rather than concentrated as total earnings loss. The assumed positive wage pass-through is a large fiscal offset in this grid. The Gini remains positive in every central-displacement wage cell, including the adverse-wage cases; more positive survivor gains are associated with a larger Gini response, consistent with a wider gap between displaced workers and those who keep their jobs.

4.7 The age and gender dimensions

The incidence axis has a within-family age structure worth drawing out. Under exposure-proportional incidence, workers aged 16–24 account for only 5.3 per cent of the displaced in the central scenario, because young people in the UK are concentrated in low-exposure service occupations; the burden falls on prime-age workers (27.1 per cent of the displaced are aged 35–44). The junior-concentrated family of Table 2 reweights displacement probabilities towards junior workers while holding the aggregate displacement rate at 7 per cent; because juniors are concentrated in occupation groups whose fixed quotas the tilt does not change, the realised seed-0 age composition barely moves (a 16–24 share of 4.8 per cent and a 55–64 share of 18.3

Table 4: Adjustment-margin and survivor-wage sensitivities (seed 0, 2026)

Scenario	Displaced (million)	Exchequer cost (£bn/yr)	Δ Poverty BHC (pp)	Δ Gini (pp)
Fixed gross earnings loss (7.77 per cent); λ scales displacement				
$\lambda = 0$	0.00	26.6	+0.10	−0.44
$\lambda = .25$	0.42	24.5	+0.61	−0.03
$\lambda = .50$	0.77	22.7	+1.02	+0.29
$\lambda = .75$	1.15	20.6	+1.39	+0.62
$\lambda = 1$	1.56	18.2	+1.81	+1.04
Central 7 per cent displacement with alternative survivor-wage effects				
−5.0%	1.56	57.3	+2.65	+0.59
−2.6%	1.56	45.0	+2.21	+0.73
0.0%	1.56	31.6	+2.01	+0.88
+2.6%	1.56	18.2	+1.81	+1.04
+5.0%	1.56	5.8	+1.68	+1.19

Note: In the mixed panel, λ thins the central 7 per cent employee-displacement draw and C-AIOE-graded survivor cuts fill the gap to that same seed-0 draw's gross pre-uplift earnings loss (7.77 per cent of the wage bill). Gross loss is identical in every row; the 2.6 per cent complementarity-graded uplift and capital shock then apply. The pure wage-cut endpoint therefore matches the C-AIOE wage-margin row's gross-loss calibration. The second panel holds displaced workers and the capital shock fixed and changes only the employment-weighted survivor-wage parameter. These are author-designed reduced-form comparative statics, not estimates of equilibrium wage responses.

against 18.1 per cent), and the aggregate differences are small relative to variation across the 50 assignment draws for cost (£17.9 ± 1.6 under the junior tilt against £17.7 ± 1.2 billion under exposure-proportional incidence), poverty and the Gini alike. The exercise shows that the junior tilt is an additional scenario assumption rather than a consequence of occupational exposure alone—and that, within fixed occupation quotas, it does little to redirect the shock towards the young. Even under the junior tilt, prime-age workers remain the largest displaced group.

If AI adoption in practice falls hardest on juniors through recruitment rather than dismissal—as the seniority-biased hiring evidence of Hosseini Maasoum and Lichtinger (2026a) and the early-career employment declines in AI-exposed occupations documented by Brynjolfsson et al. (2025a) suggest—the fiscal consequences could differ materially from the broad-based scenarios above, because junior workers contribute little income tax and National Insurance. That channel, however, operates through hiring flows that stock-based simulation frameworks such as mine do not capture by construction.

Exposure also carries a gender gradient. Employed women in the FRS have a mean C-AIOE of 0.33 against 0.18 for employed men, reflecting women's concentration in administrative, professional and associate-professional work. In the central scenario this translates into a displacement rate of 7.5 per cent for employed women against 6.5 per cent for men (means over 20 draws): women hold 51.8 per cent of employment but account for 55.4 per cent of the displaced. Household-level income changes are nevertheless similar by sex (−2.9 per cent for both women and men). A natural reading is that income pooling within couples mutes the individual asymmetry at the household level, though I have not decomposed the result and offer this as a hypothesis rather than a finding. The individual-level asymmetry itself echoes the international evidence that female employment is substantially more exposed to generative AI than male employment (ILO, 2025).

4.8 Benefit caseloads

The fiscal aggregates above can also be read on the delivery side, as caseloads. In the central scenario the shock draws 370,000 benefit units into Universal Credit while 59,000 exit, a net caseload increase of 311,000 benefit units, and annual UC spending rises by £2.6 billion from a baseline of £41.4 billion, of which £0.4 billion is the housing element within paid awards. Against DWP's roughly £334 billion of welfare spending in 2025–26 (a context figure, not a model output), the central increment is about 1 per cent; the high scenario roughly doubles it (674,000 net new benefit units, +£6.3 billion), while the low scenario adds only 30,000 benefit units and £0.3 billion. Caseload responses vary with incidence in the expected direction: the junior-concentrated family, whose displaced workers more often live in households with other earners, adds fewer benefit units (251,000 net) at lower cost (+£2.4 billion). Two accounting notes apply: only the benefit-unit figures are net of exits—the corresponding household and person counts (353,000 and 774,000 in the central scenario) are gross new entrants—and the housing element is measured within paid awards, slightly overstating housing-specific cash.

5 Policy responses

If the shock modelled above is structural rather than cyclical—a permanent reallocation of labour demand rather than a downturn that unwinds—then the question facing the state is not whether the existing tax-benefit system cushions the blow (Section 4 shows that it does, absorbing roughly a third of the market-income loss in the hardest-hit decile) but whether it cushions it well, and at what fiscal cost the residual hardship could be bought down. This section evaluates three stylised policy responses inside the shocked world. Each reform is simulated on top of the central scenario (7 per cent displacement, seed 0, 2026), so the estimand throughout is the difference between the shocked-plus-reform and shocked simulations: what the reform adds, given that the shock has happened. I evaluate them only as responses to the shock, not as standing policy; each is modelled for 2026 alone.

5.1 Three stylised reforms

The first reform (R1) is a wage-insurance scheme in the spirit of trade-adjustment proposals: displaced workers receive 50 per cent of their lost earnings, capped at £15,000 per year. I implement it as a post-simulation transfer that is non-taxable and disregarded by means tests, so its gross cost equals its net cost; this brackets the generous end of plausible implementations. The second (R2) is a demand-side circuit breaker within the existing architecture: a 20 per cent increase in the Universal Credit standard allowance, implemented as a parameter reform inside PolicyEngine so that all interactions with tapers, caps and passported entitlements are captured. The third (R3) suspends the benefit cap and cuts the UC earnings taper from 55 to 45 per cent, targeting the households for whom the existing system's own restraints bind hardest. R2 and R3 costs are net of clawback through the rest of the system.

Table 5 reports the results and Figure 9 plots the decile pattern of the gains. The table reports simulated fiscal cost per poverty point averted; it is not a complete welfare or policy appraisal. The UC uplift averts a percentage point of after-housing-costs poverty for £9.9 billion, and the cap-and-taper package for £9.0 billion, against £13.8 billion for wage insurance. This

Table 5: Policy reforms on top of the central shock, 2026 (exposure-proportional incidence, seed 0)

Reform	Annual cost (£bn)	Poverty change (pp) BHC	AHC	Gini change (pp)	Cost per pp averted (£bn) BHC	AHC
R1 Wage insurance	20.6	−1.41	−1.49	−0.92	14.6	13.8
R2 UC allowance +20%	5.7	−0.71	−0.58	−0.30	8.1	9.9
R3 Cap suspension + taper cut	4.6	−0.46	−0.51	−0.22	9.9	9.0

Note: All effects are shocked-plus-reform minus shocked, fixed seed-0 draw, reforms in force for 2026 only. R1: 50 per cent of lost earnings, £15,000 cap, paid to 1.56 million displaced workers (mean transfer £13,252), modelled as non-taxable and means-test-disregarded so gross cost equals net cost. R2 and R3 are PolicyEngine parameter reforms; costs are net of clawback. R3 combines benefit-cap suspension with a UC taper cut from 55 to 45 per cent. Poverty lines are held at their shocked-world values.

pattern reflects the specifications: wage insurance is earnings-proportional, and because the exposure-proportional shock displaces workers who are disproportionately in the upper half of the distribution, roughly 55 per cent of R1 reform pounds accrue to households in baseline deciles 8–10. The UC-side instruments, by contrast, concentrate 71 per cent (R2) and 64 per cent (R3) of reform pounds in deciles 1–5. Each household's total gain is counted once with its household weight; it is not broadcast and re-counted for every member. Wage insurance does buy roughly twice the total poverty reduction of either UC reform—it is by far the largest intervention—but at roughly three-and-a-half to four times the cost, and with a larger simulated cost per poverty point.

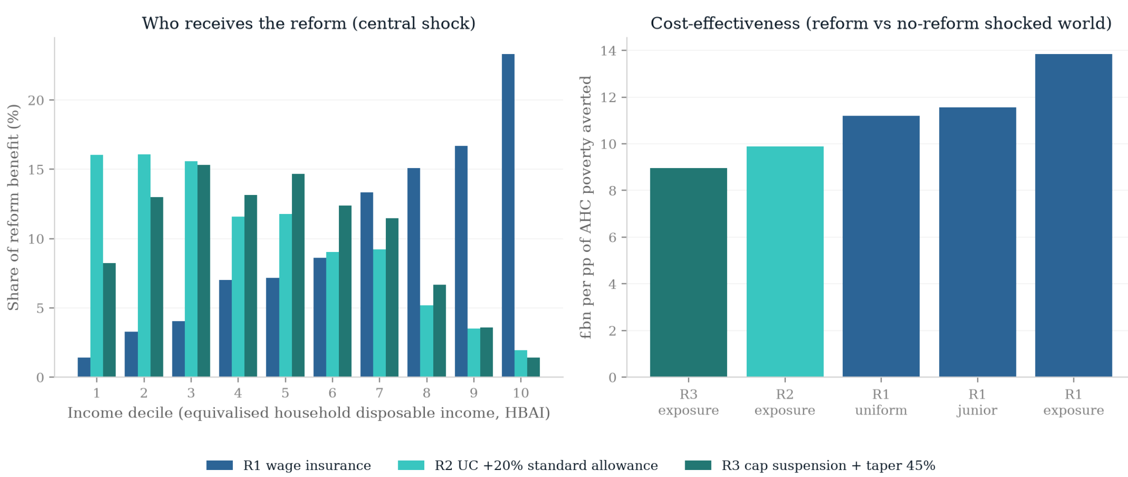

Figure 9: The three policy reforms on top of the central shock: fiscal cost, poverty and Gini effects, and the decile distribution of household gains. All panels show shocked-plus-reform minus shocked, seed 0, 2026.

The reported reform ratios are less sensitive to the displacement draw than the shock aggregates are: over 50 Monte Carlo draws the wage-insurance cost varies by only ±£0.5 billion (£21.4 ± 0.51 billion under exposure incidence, averting 1.52 ± 0.10 BHC points), and the UC-side instruments by less (R2: £5.7 ± 0.04 billion, −0.67 ± 0.04 points; R3: £4.6 ± 0.06 billion, −0.47 ± 0.04 points), and the ordering of the reported cost-per-point ratios is unchanged across these assignment draws. The ordering is also unchanged in the 50 per cent earnings-retention hybrid of Section 4.3: with displaced workers retaining half a year's earnings (status and hours still full-year unemployed), a duration-consistent wage insurance costs £30 $\pm$ 3 billion per BHC poverty point averted against £8.5 $\pm$ 0.2 billion for the UC uplift, because shortening the earnings loss shrinks the poverty problem faster than it shrinks the earnings-proportional transfer. Because incidence is a central scenario uncertainty, I re-run the wage-insurance reform under the junior-concentrated and uniform incidence families. Within the two additional incidence variants, R1 costs £11.6 billion per AHC point under junior incidence and £11.2 billion under uniform incidence, compared with the £9.0–9.9 billion achieved by the UC-side instruments under the central family. The gross cost of R1 is nearly invariant across families (£19.9–20.6 billion), consistent with the aggregate displaced wage bill being held approximately fixed; what varies is how much of that spending lands near the poverty line.

None of this settles the choice of instrument. Wage insurance addresses consumption smoothing and earnings-loss compensation for displaced workers, whereas the UC reforms target low household income. Under the simulated designs and using poverty averted per pound as the sole criterion, the two UC reforms have lower cost-per-point ratios than this wage-insurance design. The comparison does not rank the instruments on insurance, consumption smoothing, administration, financing or wider welfare objectives.

5.2 The tax-composition channel

The reforms above add expenditure after the shock has weakened the simulated fiscal position, and the size of that damage depends on a question the microsimulation cannot answer by itself: where does the displaced wage bill go? The March 2026 Economic and Fiscal Outlook does not model AI displacement. It does, however, note more generally that a shift in income growth from labour towards profits depresses receipts because labour income carries a higher effective tax rate (OBR, 2026). I use that tax-composition mechanism only as motivation and price it with an author-designed accounting sensitivity rather than presenting it as an OBR AI scenario. Let ϕ denote the author-chosen share of the displaced wage bill that reappears as corporate profits taxed at the 25 per cent main rate. This exercise runs on the displacement-only, narrow-tax baseline: it prices the displaced wage bill and applies no wage uplift, so the £21.2 billion $\phi = 0$ shortfall below is the 7 per cent/no-uplift cash-tax cell of Section 4.4, not the £18.2 billion full-balance central. On that baseline the displaced wage bill is £78.0 billion and the effective income-tax-plus-NICs rate on displaced workers' earnings is 25.6 per cent, so the labour tax forgone is £20.0 billion—and because that effective labour rate almost exactly equals the corporation-tax main rate, the shortfall is very nearly $(1-\phi)\times 0.25\times$ the displaced wage bill. At $\phi = 0$ the net revenue shortfall is the microsimulation's £21.2 billion; at $\phi = 0.5$ it falls to £11.4 billion; and at $\phi = 1$ the labour-to-capital channel is almost exactly self-financing (a £0.5 billion shortfall relative to full labour taxation), leaving only a residual £1.7 billion net cost, chiefly shock-induced benefit spending together with smaller non-labour-tax revenue changes. Figure 10 traces the full grid over displacement rates and ϕ.

Revenue neutrality is not necessarily distributional neutrality. At $\phi = 0.5$ I simulate an author-chosen payout ratio of 0.5: £14.6 billion of post-corporation-tax profits are distributed pro rata to existing FRS dividend holders, with dividend tax captured in the simulation. The Gini rises by 0.46 percentage points relative to the shocked world without recycling. Both poverty measures change by less than displayed precision, and no dividend gains are recorded in deciles 1–4. Because dividend records are sparse and top incomes are under-represented in

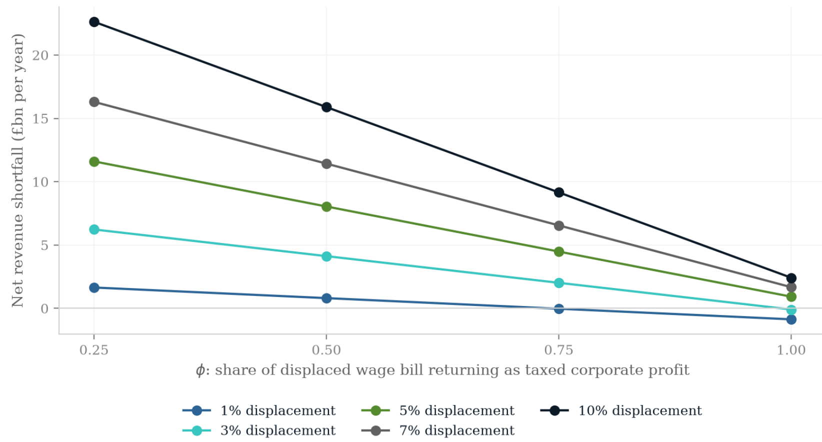

Figure 10: Net revenue change by displacement rate and ϕ, the share of the displaced wage bill reappearing as corporate profits taxed at 25 per cent. The $\phi = 0$ line is the microsimulation's net revenue change; higher ϕ adds recouped corporation tax outside the simulation.

the FRS, these exact zeros should not be interpreted as population estimates. The exercise illustrates how, under this particular allocation rule, substantial tax recovery can coexist with a more unequal household-income distribution.

Two caveats bound this exercise. It is static accounting layered on top of the microsimulation: the incidence of corporation tax is not modelled, the corporation-tax base is assumed equal to accounting profit (no allowances, losses or profit shifting), and the effective labour rate is a pro-rata attribution of income tax and NICs to employment income that ignores the schedule ordering of savings and dividend bands. And ϕ itself is the free parameter the whole channel turns on; I present a grid rather than a point estimate because nothing in my framework pins it down. Relatedly, the $\phi = 1$ result is closer to an accounting identity than an empirical finding; the object of interest is the distributional consequence at every ϕ, not the revenue arithmetic at the endpoint.

6 Conclusions and policy implications

The results should be read as scenario analysis, not prediction. I do not forecast the pace or extent of AI adoption in the United Kingdom; I trace the distributional and fiscal consequences that would follow *if* labour-market shocks of a given size and occupational incidence were to materialise, under tax-benefit rules as legislated. The observational record remains too thin to adjudicate between the incidence families: the earliest evidence ranges from minimal measured earnings and hours effects to date at one extreme to a modest relative contraction in exposed occupations at the other, none yet at the scale of my central shock. That unsettled state is the case for spanning the disagreement rather than resolving it by assumption, and what the paper contributes is not a claim about which world will arrive but a quantification of what turns on the answer.

The results can be benchmarked against the one directly comparable exercise, the Irish tax-benefit simulation of Doorley et al. (2026), which uses the same central conventions (7 per cent displacement, 2.6 per cent survivor-wage growth, +0.4 points on the return to capital) on

Irish SILC data. Table 6 compares the two. The qualitative pattern transfers: displacement concentrates in upper-income households, the tax-benefit system absorbs a substantial share of the loss, and measured inequality rises in every scenario in both countries. The main quantitative difference—greater absorption at the top in Ireland—is consistent with the two systems' relative progressivity, and the fiscal magnitudes are not directly comparable because Doorley et al. (2026) report Exchequer effects only relative to a net-of-welfare revenue base.

Table 6: Comparison with the Irish simulation of Doorley et al. (2026)

	Ireland (SWITCH, SILC)	UK (this paper)
Central shock conventions	7% / +2.6% / +0.4pp	identical
Incidence of displacement	concentrated in upper deciles	0.5% (bottom) to 4.9% (top decile)
Gini response	rises in every scenario	rises in every family and grid cell
Absorption of top-decile loss	roughly half	roughly one third
Exchequer effect (worst grid cell)	$\approx -25\%$ of net revenue base	-9.4% of gross simulated IT+NI

Note: Irish entries summarise the published report, which states fiscal results only as percentages of net government revenue from income tax, social security and welfare; the revenue bases therefore differ by construction and the last row is not a like-for-like comparison. Both exercises are conditional simulations on different microdata, occupational resolutions and tax-benefit systems.

Two structural features give the simulated displacement results their shape. Means-tested support absorbs losses low in the distribution and progressive taxation forgoes revenue higher up, so household cushioning appears as a deterioration in the public finances. Inequality rises because zero earnings for selected workers, partial tax-benefit replacement, gains for survivors and the capital uplift jointly widen post-shock income gaps. This explanation is conditional on the simulated channels; it is not caused mechanically by the Gini "re-ranking" households.

6.1 Policy implications: what depends on who bears the shock

The central policy implication is conditional on incidence. Measured inequality rises in each simulated displacement family, while the balance between fiscal and poverty effects varies across the specified allocations.

If the shock is top-loaded, as in the exposure-proportional family, the simulated fiscal effect is relatively large. Displaced workers are drawn from occupations that pay well and are taxed heavily, so each displacement removes income tax and National Insurance while adding comparatively little to means-tested expenditure. The Exchequer cost of the central shock is £17.7 billion under exposure-proportional incidence against £13.9 billion under uniform incidence (50-draw means)—the same aggregate shock, roughly a quarter more expensive under the different incidence assignments. Under top-loaded incidence one policy concern is revenue resilience: the UK tax base leans heavily on the taxation of labour income, and the workers most exposed are precisely those who contribute most to it. If AI additionally shifts the functional distribution of income from labour towards capital, the base that finances the cushioning I document would itself erode—a possibility that has motivated recent work on the taxation of capital in an AI-intensive economy (Bastani and Waldenström, 2024). The tax-composition exercise of Section 5 prices that possibility for the Treasury: whether base erosion is a first-order fiscal problem is highly sensitive to how much of the displaced wage bill reappears as taxable corporate profit.

Under flatter or junior-concentrated incidence, the mean fiscal effect is smaller but the mean poverty effect is not. In the paired draw-level comparisons of Section 4.5, the junior-concentrated fiscal cost is statistically indistinguishable from the exposure-proportional one, and while the uniform family's Gini effect reliably exceeds the exposure-proportional one, its poverty effect does not; these are scenario comparisons across shared assignment draws, not population-inference tests. These scenarios make out-of-work support and labour-market entry relevant policy concerns. One limitation deserves emphasis. The strongest early empirical evidence locates AI's employment effects among junior workers *within* exposed occupations and firms—relative employment declines of about 13 per cent for the youngest workers in the most exposed US occupations (Brynjolfsson et al., 2025a) and UK job cuts concentrated almost entirely in junior roles (Klein Teeselink, 2025; Hosseini Maasoum and Lichtinger, 2026a)—operating chiefly through reduced hiring. A stock-based, occupation-proportional simulation cannot capture that margin by construction: allocating displacement across occupations makes 16–24-year-olds only about 5 per cent of the displaced, because they are under-represented in high-exposure professional occupations. The framework therefore likely understates youth incidence. Because junior workers earn less, correcting this would if anything tilt losses further down the age-earnings profile and push measured inequality up, so the headline sign of the Gini result is robust to it; but the fiscal cost would fall, since juniors contribute less income tax. My one-year reweighting exercise also does not quantify dynamic scarring or persistent earnings effects of delayed entry.

For the simulated designs, the UC-side reforms avert more poverty per pound than the wage-insurance specification. This supports considering existing means-tested mechanisms when poverty reduction is the objective, but it does not establish an overall instrument ranking: wage insurance serves different insurance and consumption-smoothing aims, and administrative, behavioural and financing effects are outside the comparison.

The expertise-compression and Klein-anchored families combine relatively large fiscal and poverty effects within the author-chosen calibrations. This is compatible with Hosseini Maasoum and Lichtinger (2026b), whose labour-supply channel concerns between-occupation wages rather than the household displacement exercise here. In the separate wage-margin sensitivity, my author-imposed PSS-proportional schedule raises the fiscal cost to £26.4 billion, largely removes the poverty increase and turns the Gini change slightly negative. These results show how outcomes vary across the schedules simulated here; they do not establish policy dominance or general upper and lower bounds.

Two considerations recur across the simulated displacement families. First, targeted retraining is one relevant lever: Hyman et al. (2025) report rising returns to publicly funded retraining for workers from highly AI-exposed occupations in the United States, largely through moves into less-exposed work. Whether this finding transports to the UK is not tested here. Second, measured inequality rises in every displacement family examined, which motivates monitoring; this is a conditional model result rather than a forecast.

Across the simulated displacement scenarios, the tax-benefit system partly cushions market-income losses at an incidence-dependent fiscal cost. The results point to three areas for further analysis rather than a settled policy prescription: occupational reallocation, entry prospects for younger workers, and the resilience of a labour-income-heavy tax base.

6.2 Limitations and further work

Several limitations qualify the results; I state them by pipeline stage, with the measurement conventions needed to compare my figures with official statistics.

The first stage is the data and its weighting. All results use the HBAI equivalised household disposable income concept, with the Gini computed after bottom-coding negative incomes at zero. My baseline Gini of 0.309 sits below the official 0.34 because I use the plain Family Resources Survey without the Survey of Personal Incomes (SPI) top-income adjustment, so levels are not comparable to official statistics even though the changes I report are computed on a consistent basis. I use the uncalibrated FRS grossing weights rather than PolicyEngine's enhanced dataset, whose household cloning would break the record-to-occupation correspondence the exposure join depends on; aggregate fiscal magnitudes therefore carry the usual survey-grossing caveats. Single-draw figures also carry displacement-draw noise that the 20-draw averages remove (seed conventions in Section 3.5 and Appendix A.1).

Occupational exposure is assigned at the 1-digit SOC level, because the FRS records occupation only at the major-group level, so within-group heterogeneity in exposure (Felten et al., 2021; Webb, 2019; Eloundou et al., 2024) is averaged away. The scores also reach the FRS through a reconstructed pipeline—a rebuilt complementarity adjustment and a chained SOC2018–SOC2010–ISCO–SOC2020 crosswalk with ASHE weights for 340 of 412 unit groups. Adopting the UK-native GAISI index (Henseke et al., 2025) together with detailed-occupation imputation from the Labour Force Survey would remove both the crosswalk and the resolution constraint, and is my first priority; for the expertise-compression family specifically, the public occupation-level expertise measures of Hosseini Maasoum and Lichtinger (2026b) are already used as the PSS gradient of the wage-margin family (Section 4.6), and their finer skill-specificity measures offer a further robustness input.

The displacement treatment is static: displaced workers are fully out of work for the simulated year (earnings, hours, pension contributions and salary sacrifice zeroed, entitlements recomputed), while dual jobholders keep self-employment income. Unemployment duration is not modelled, so contributory-benefit exhaustion (new-style Jobseeker's Allowance lasts at most six months) is not captured, and over longer horizons I overstate support for displaced workers without means-tested entitlements. The sensitivity runs of Section 4.3 indicate substantial sensitivity to duration—the 50 per cent earnings-retention hybrid, which halves annual earnings while leaving status and hours at full-year unemployment, cuts the fiscal cost by more than three-quarters—while the single 70 per cent take-up sensitivity changes the headline estimates little. The wage-margin and mixed-adjustment exercises of Section 4.6 relax the polar assumption that adjustment is either full job loss or pure wage cuts: holding gross pre-uplift earnings loss fixed at the paired draw's realised value, the fiscal cost falls from £26.6 to £18.2 billion as adjustment shifts from wage cuts to displacement, while poverty and inequality rise towards their displacement endpoints. Allowing the workers who retain jobs to lose wages rather than receive the assumed productivity uplift is more demanding: at fixed 7 per cent displacement, a 2.6 per cent survivor-wage decline raises the fiscal cost to £45.0 billion and BHC poverty to +2.21 points. Ripple effects onto unexposed occupations are separately priced in reduced form in Appendix A.8: routing displaced labour supply to destination occupations and cutting non-displaced wages by an assumed inverse-demand slope of 0.3 raises the central fiscal cost to £28.0 billion. The fiscal estimate rises under the adverse survivor-wage and selected ripple specifications, but because the slope is assumed rather than estimated, no conclusion that the central scenario is conservative follows; these exercises do not bound other labour-market responses.

The capital channel is narrow: it covers only interest and dividends, excluding rental income and returns inside private pensions, and surveys under-represent top and capital incomes, so its top-decile concentration may be understated, although neither the direction nor magnitude of that bias is identified here. The channel, and the fiscal aggregates more broadly, are also sensitive to the choice of dataset: rerunning the central scenario on PolicyEngine's enhanced FRS dataset raises the seed-0 Exchequer cost from £18.2 billion to £29.5 billion and the BHC poverty rise from +1.81 to +3.0 points, with 2.97 million rather than 1.56 million workers displaced at the same 7 per cent rate, because the enhanced dataset's calibrated weights and incomes gross to a substantially different employee base (Appendix A.10). Headline fiscal magnitudes should therefore be read as conditional on the plain-FRS base. Linked wealth data (the Wealth and Assets Survey) would sharpen it.

All scenarios exclude the self-employed, over whom the exposure indices and displacement literature are not defined; since self-employment is a meaningful, partly exposed share of UK employment, this is a substantive omission.

Finally, the mechanisms invoked for some decile patterns—including savings buffers and income pooling within couples—are hypotheses consistent with the data, not formal decompositions. Across the scenarios examined, the tax-benefit system partly cushions household losses, and the extent of that cushioning depends on shock size, incidence and adjustment margin.

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A Additional tables and figures

A.1 Seed and draw conventions

Displacement is a random draw of which exposed workers lose employment, subject to occupation-group quotas proportional to employment times mean exposure. Throughout the paper, three conventions apply. The scenario grid (Figures 5–7, and their appendix variants below), the decomposition in Figure 4 and the preset scenarios are single draws at seed 0, while the five incidence families report means over 50 draws (seeds 0–49). The decile transition and wage-gain profiles (Figures 1 and 2) are means over the same 50 paired draws. The robustness summary and the assignment-draw intervals in Figure 18 are 20-draw Monte Carlo runs, with ± values reported as standard deviations across draws. Within each draw, ordering keys are uniform within occupation group, so a record's inclusion probability does not depend on its grossing weight.

A.2 Job loss by occupation group

Table 7 reports the central-scenario displacement rate for each SOC2020 major group. The allocation rule concentrates losses in administrative and secretarial work (10.5 per cent of the group's employment), managerial and professional occupations (9.2–9.7 per cent) and associate professional roles, while elementary occupations—the least-exposed group, whose normalised exposure score is zero—experience no displacement at all. Clerical and professional work bears the shock; manual and elementary work is insulated.

Table 7: Central-scenario job loss by SOC2020 major group

Major group	C-AIOE	Employment (m)	Job loss (%)
1 Managers, directors and senior officials	0.62	2.45	9.7
2 Professional occupations	0.60	6.46	9.2
3 Associate professional occupations	0.47	3.13	7.7
4 Administrative and secretarial occupations	0.74	2.35	10.5
5 Skilled trades occupations	−0.33	1.56	2.7
6 Caring, leisure and other service occupations	−0.14	2.32	3.6
7 Sales and customer service occupations	0.27	1.22	7.4
8 Process, plant and machine operatives	−0.54	1.41	1.3
9 Elementary occupations	−0.73	2.05	0.0

C-AIOE is the standardised complementarity-adjusted exposure score before the min-zero normalisation used in the allocation rule. Employment and job-loss rates are survey-weighted; single draw, seed 0.

A.3 Where the shock lands: baseline distributions

Figures 11–15 show the baseline household-level decile distributions of the income components through which the three channels operate (aggregate £billion per year by decile of equivalised household disposable income). Market income excluding capital (£1,313 billion in total) and income tax and National Insurance (£312 billion) are heavily concentrated in the top deciles—which is why displacement of high earners is fiscally expensive. Benefits (£320 billion) are spread across the lower and middle deciles, peaking in deciles 3 to 5. Capital income (£23 billion of interest and dividends) is the most concentrated series of all: the top decile holds more than a third of the total.

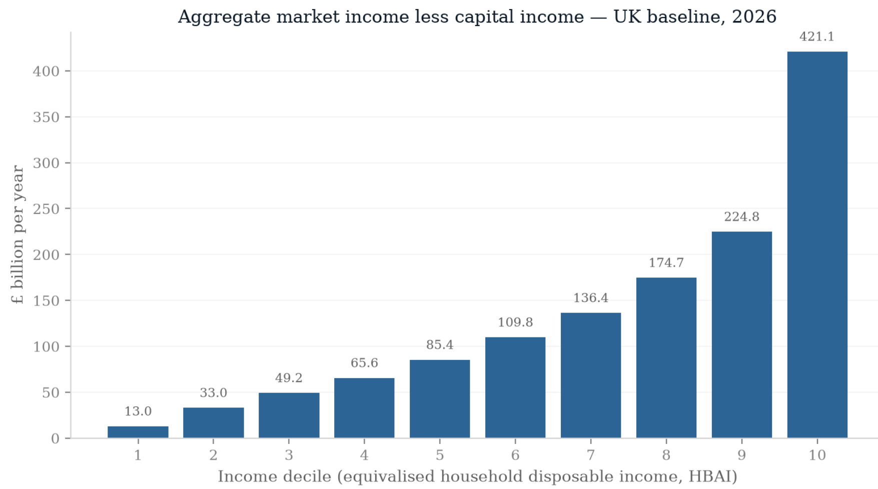

Figure 11: Baseline aggregate household market income (less capital income) by decile.

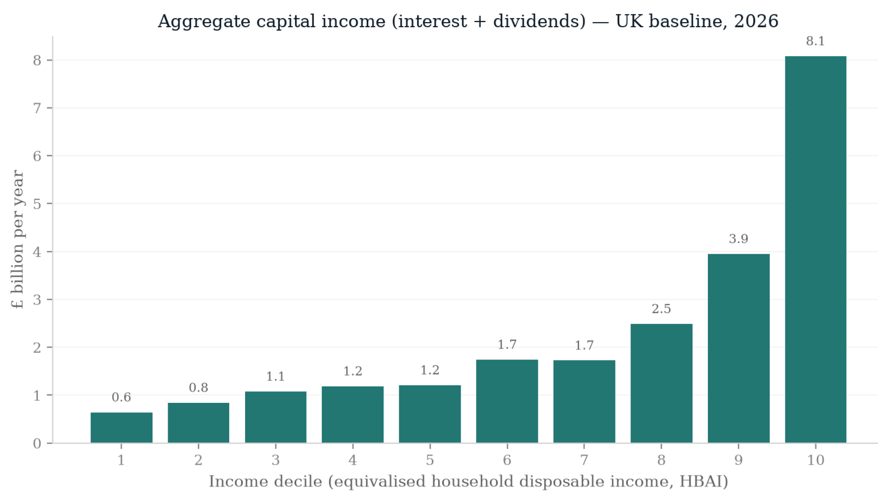

Figure 12: Baseline aggregate household capital income (interest and dividends) by decile.

A.4 Exposure incidence before any simulation

Figures 16 and 17 show, without any shock, how workers in each income decile distribute across employment-weighted tertiles of AI exposure and of complementarity. The share of workers in the top exposure tertile rises steadily with household income, while complementarity shows no comparable gradient—the descriptive facts that generate the simulated incidence in Section 4.

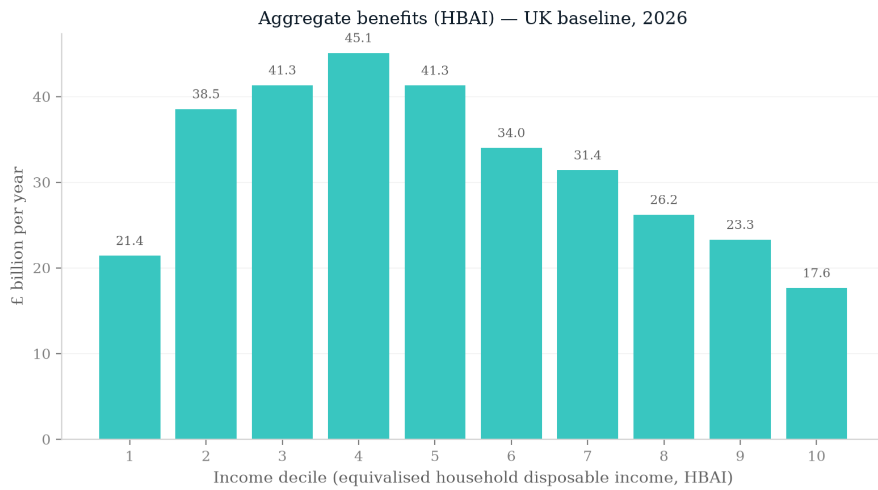

Figure 13: Baseline aggregate household benefits by decile.

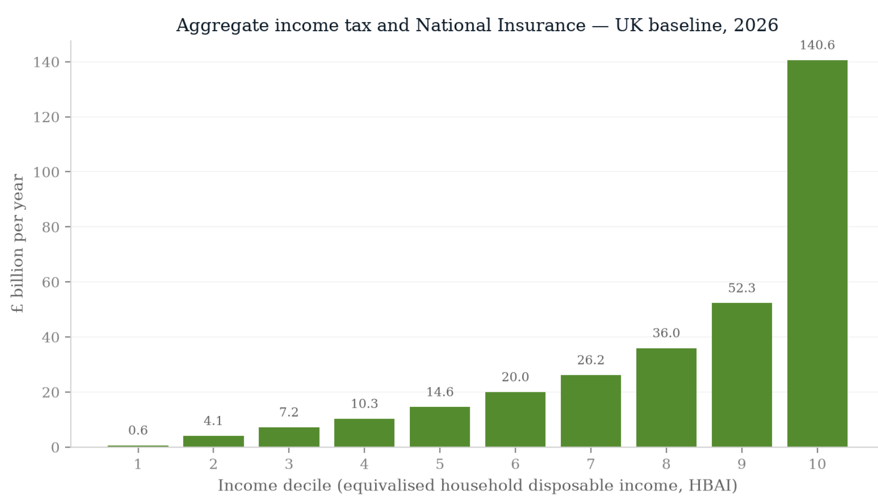

Figure 14: Baseline aggregate household income tax and National Insurance by decile.

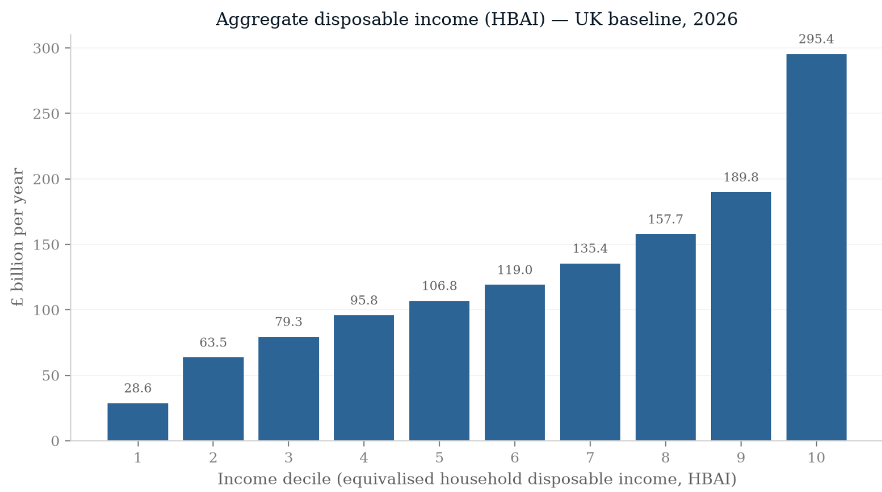

Figure 15: Baseline aggregate household disposable income (HBAI concept) by decile.

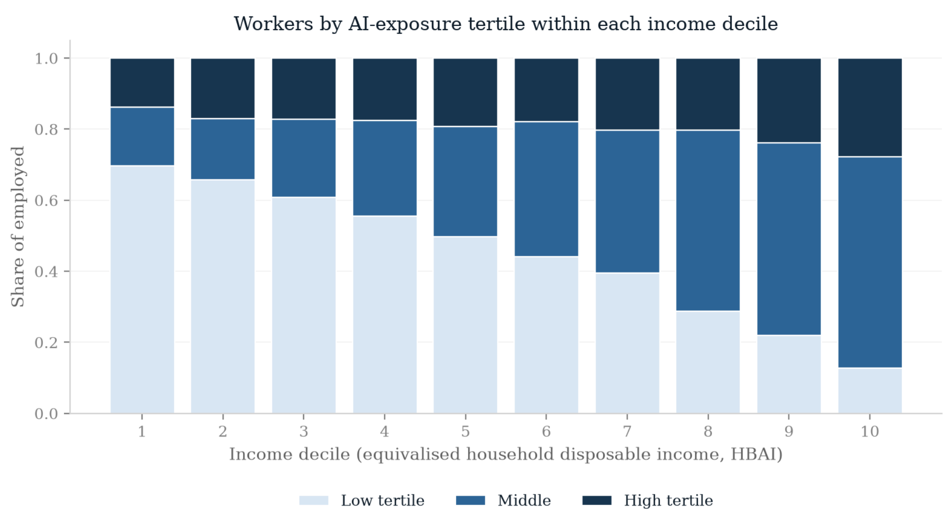

Figure 16: Distribution of employed persons across AI-exposure tertiles, by income decile.

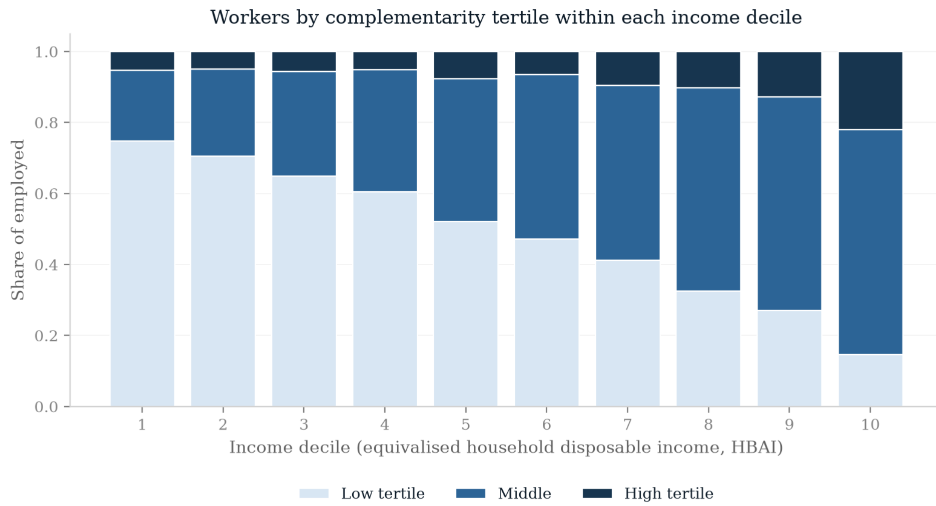

Figure 17: Distribution of employed persons across complementarity tertiles, by income decile.

A.5 Monte Carlo uncertainty on the decile profile

Figure 18 reports the central-scenario disposable-income change by decile as the mean over twenty independent displacement draws with central 95 per cent intervals across assignment draws. The gradient runs from −0.75 per cent in the bottom decile to −3.87 per cent in the top. On the current 20-draw means it is not everywhere strictly monotone: decile 3 loses slightly less than decile 2 (−0.85 against −0.93 per cent) and deciles 9 and 10 are essentially tied (−3.87 per cent each), with both differences well inside the draw noise. The overall upward slope of the loss profile appears across the reported draw interval.

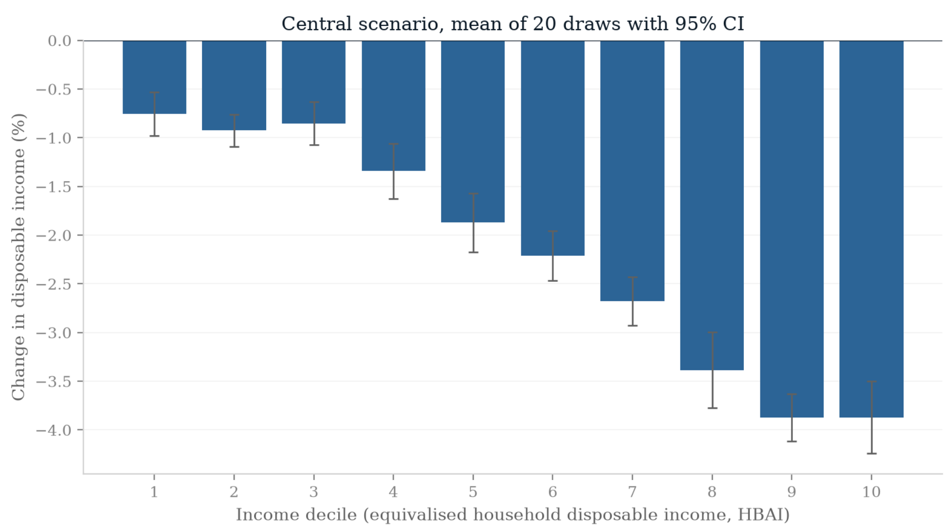

Figure 18: Disposable income change by decile, central scenario: mean and central 95 per cent interval across 20 assignment draws.

A.6 Uniform-shock and alternative-index comparisons by decile

Figure 19 compares the central scenario against the same aggregate shocks allocated uniformly across occupations, decile by decile; Figure 20 repeats the central scenario with the Eloundou et al. (2024) GPT-exposure index in place of the C-AIOE. The exposure-based allocation shifts losses up the distribution relative to a uniform shock (top-decile losses of 5.3 against 2.5 per cent in this single seed-0 draw); lower-decile comparisons are noisy at the single-draw level, and the choice of exposure index leaves the decile profile broadly similar.

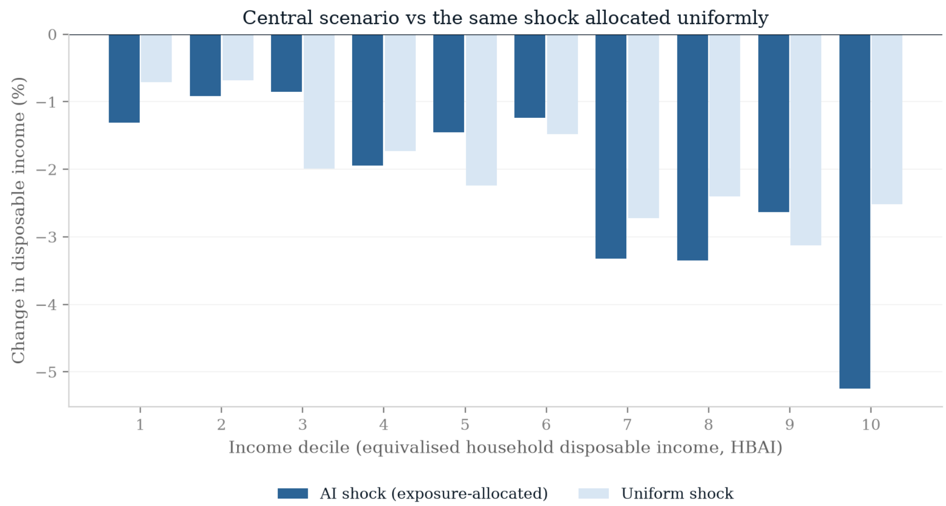

Figure 19: Disposable income change by decile: exposure-allocated versus uniform shock, central parameters.

A.7 Full exposure-index sensitivity

Figure 20 compared decile profiles under two indices; here I rerun the entire central scenario under five: the C-AIOE (my central measure), the unadjusted Felten et al. (2021) AIOE, the Eloundou et al. (2024) GPT-exposure beta, and the two indices from the UK government's DSIT exposure assessment (all-AI and large-language-model variants). The comparison matters because the published literature disagrees sharply on *levels*: estimates of the share of employment "highly exposed" to AI range from a few per cent to well over half depending on the index and threshold chosen, so any result that survived under only one index would be fragile. My quota mechanism holds the aggregate displacement rate at 7 per cent, so the indices differ only in *where* the same aggregate shock lands across occupation groups.

Table 8 and Figure 21 report the outcomes. The Exchequer cost spans £16.7–20.4 billion (the largest deviation from the C-AIOE figure is +11.9 per cent, under the Eloundou GPT-exposure beta), the Gini change spans +0.94 to +1.18 points, and the before-housing-costs poverty change spans +1.81 to +2.04 points. The decile-10 transition share exceeds the decile-1 share by a factor of 8.4 to 15.0. Within this seed-0 sensitivity, all five indices produce a positive Gini change, a transition gradient that rises with income, and a fiscal cost within 20 per cent of the C-AIOE result. This is consistent with their similar major-group rankings and the fixed aggregate quota.

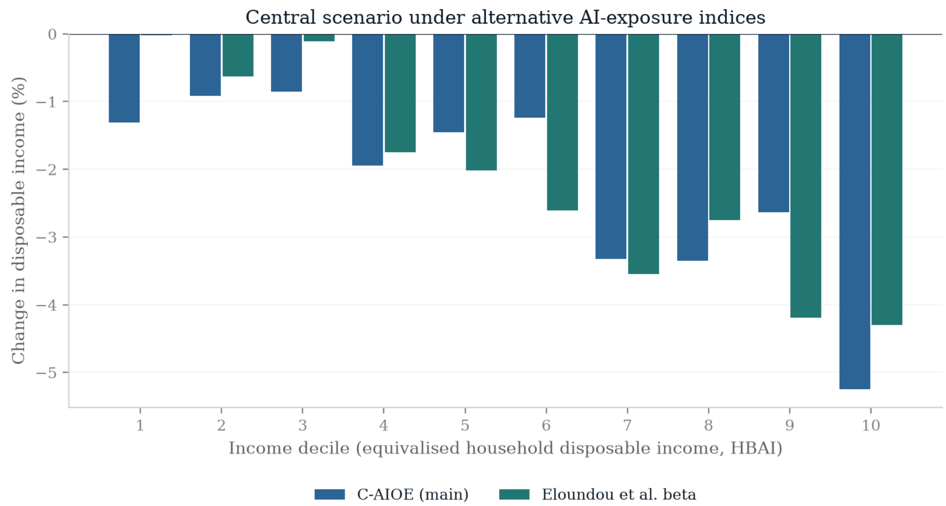

Figure 20: Disposable income change by decile under the C-AIOE and the Eloundou et al. exposure index.

Table 8: Central scenario under five exposure indices

Index	Cost (£bn)	ΔPov. BHC (pp)	ΔGini (pp)	D1 share (%)	D10 share (%)
C-AIOE (central)	18.2	1.81	1.04	0.64	5.34
Felten AIOE	18.1	1.84	1.05	0.50	5.44
Eloundou beta	20.4	2.03	1.10	0.32	4.79
DSIT AIOE	18.2	2.04	1.18	0.36	4.89
DSIT LLM	16.7	1.83	0.94	0.43	5.80

Single draw, seed 0, 2026. D1 and D10 shares are the percentages of persons in the bottom and top baseline income deciles who transition to unemployment (displaced shares) under the draw. Aggregate displacement is held at 7 per cent by the quota mechanism under every index.

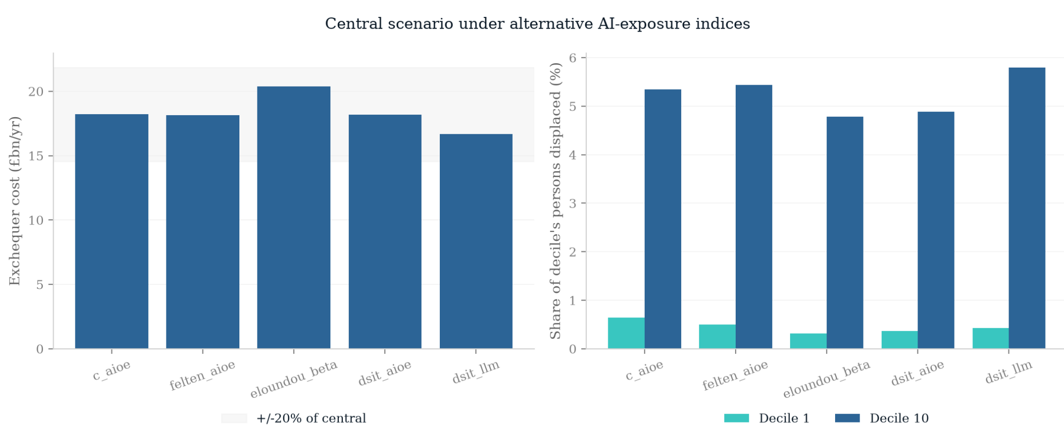

Figure 21: Headline outcomes of the central scenario under five exposure indices.

A.8 Ripple-effect sensitivity

The displacement scenarios of the main text hold non-displaced wages at the uplifted baseline, so the ripple effects of Acemoglu and Restrepo (2022)—displaced workers bidding down the wages of the non-displaced—are absent. Here I price that channel in reduced form. After the central displacement draw, each displaced worker's labour supply is routed across destination SOC2020 major groups using a row-stochastic 9×9 routing matrix with a zero diagonal: the preferred source for this matrix, a pairwise retraining-cost matrix from the Hosseini Maasoum and Lichtinger (2026b) data release, does not exist (the release contains only occupation-level measures, with no origin–destination pairs), so I fall back on employment-proportional routing, with each origin row proportional to destination employment shares from the same ASHE 2025 employment weights used throughout the paper. For each destination group I compute the proportional labour-supply inflow s_l (routed displaced FTE over baseline employment), and every non-displaced employee in group l takes a wage cut of $\eta \times s_l$ of baseline earnings, composed additively with the +2.6 per cent uplift. The parameter η is an *assumed inverse-demand slope*—the proportional wage decline per unit proportional labour-supply inflow—not a sourced own-wage elasticity, and no conclusion about whether the central scenario is conservative should be drawn from any single value of it; with 7 per cent of workers displaced the inflow shares are around 7 per cent, so $\eta = 0.3$ implies wage cuts of roughly 2 per cent across the non-displaced workforce. This is a directional sensitivity, not an implementation of the Acemoglu and Restrepo (2022) propagation matrix: there is no task structure, no equilibrium feedback, and the routing is exogenous. Table 9 reports the results for $\eta \in \{0, 0.15, 0.3, 0.5\}$.

Table 9: Central scenario with reduced-form ripple effects

η	Cost (£bn)	ΔPov. BHC (pp)	ΔPov. AHC (pp)	ΔGini (pp)
0 (central)	18.2	1.81	1.85	1.04
0.15	23.1	1.90	1.99	0.99
0.30	28.0	1.95	2.08	0.95
0.50	34.4	2.11	2.25	0.89

Single draw, seed 0, 2026. η is the author-chosen wage sensitivity applied to each destination group's proportional labour-supply inflow. $\eta = 0$ reproduces the central scenario exactly.

Under this specification, the ripple channel loads mainly on the Exchequer. At $\eta = 0.3$ the fiscal cost rises by more than half, from £18.2 to £28.0 billion, because thin, widespread wage cuts across the non-displaced workforce forgo income tax and National Insurance at the margin. The distributional estimates move less: BHC poverty rises by only 0.15 points more than in the central run, and the Gini increase falls slightly; the cuts are imposed on employed workers concentrated in the middle and upper deciles. Adding this particular ripple raises the simulated fiscal cost; other routing and wage-response assumptions could produce different results.

A.9 A historical benchmark: a 2008–09-sized employment shock

As a historical benchmark, I push a 2008–09-sized employment shock through the same framework: a uniform-incidence 2.6 per cent displacement of employees, matching the observed 2008–2010 rise in the LFS unemployment rate, with no wage or capital channel (means over five assignment seeds). The simulation yields a +0.81-point rise in absolute BHC poverty, +0.86 after housing costs, +0.29 on the relative (contemporaneous-median) line and +0.38 points on the Gini. Recorded HBAI outturns over 2007/08–2010/11 were very different: absolute BHC poverty and the Gini were roughly flat to slightly falling, and relative poverty fell outright, because benefits were uprated with high inflation while median incomes fell, and because unemployment spells were of mixed duration rather than full-year. The gap—roughly one and a half to two points of poverty per recession-sized shock—quantifies what the channels this framework deliberately excludes (benefit and threshold uprating, duration composition, behavioural and macro-policy responses) absorbed in practice. The simulated estimates throughout the paper should accordingly be read as first-year mechanical impact effects against a frozen policy baseline: an upper envelope for measured poverty responses, not a forecast of HBAI outturns. Scaled per percentage point of uniform displacement, the backcast (+0.31 points BHC per point) sits on the same ray as the paper's uniform 7 per cent family (+0.27 per point), so the benchmark and the headline results are internally consistent.

A.10 Enhanced-dataset sensitivity

On PolicyEngine's enhanced FRS 2023–24 dataset the national BHC poverty response to the central scenario is +3.0 percentage points, roughly 1.6 times the +1.81 points of the plain-FRS central run. A decomposition of the gap attributes essentially all of it to the dataset itself (+1.16 of the +1.14-point total, evaluated at the main analysis's period), with a small negative period component on the enhanced data (−0.02 points). The enhanced dataset's calibrated weights and incomes gross to a substantially different employee base (2.97 million rather than 1.56 million workers displaced at the same 7 per cent rate), so its fiscal aggregates are not comparable with the plain-FRS numbers elsewhere in the paper.

A.11 The scenario grid by decile, and without the capital shock

Figure 22 facets the full 66-cell scenario grid by baseline income decile; cells report the change in mean household disposable income (HBAI concept), which across the grid as a whole ranges from +3.0 to −5.1 per cent. Every decile's panel shifts towards losses as displacement rises, but the top deciles darken much faster: in the harshest cell (10 per cent displacement, no wage gain) the bottom decile loses 0.9 per cent of disposable income while the top decile loses 7.0 per cent. Figure 23 removes the capital-return shock. Average disposable incomes fall relative to the capital-on grid, confirming that the capital channel raises average incomes while concentrating its gains at the top. Within this grid, without the capital shock the Gini of equivalised disposable income rises in 65 of the 66 cells, by up to 1.57 percentage points; the remaining cell is the null $(0, 0)$ cell, which with the capital shock also off contains no shock at all and is exactly zero. Thus the positive Gini change in the shocked cells is not generated solely by the capital channel.

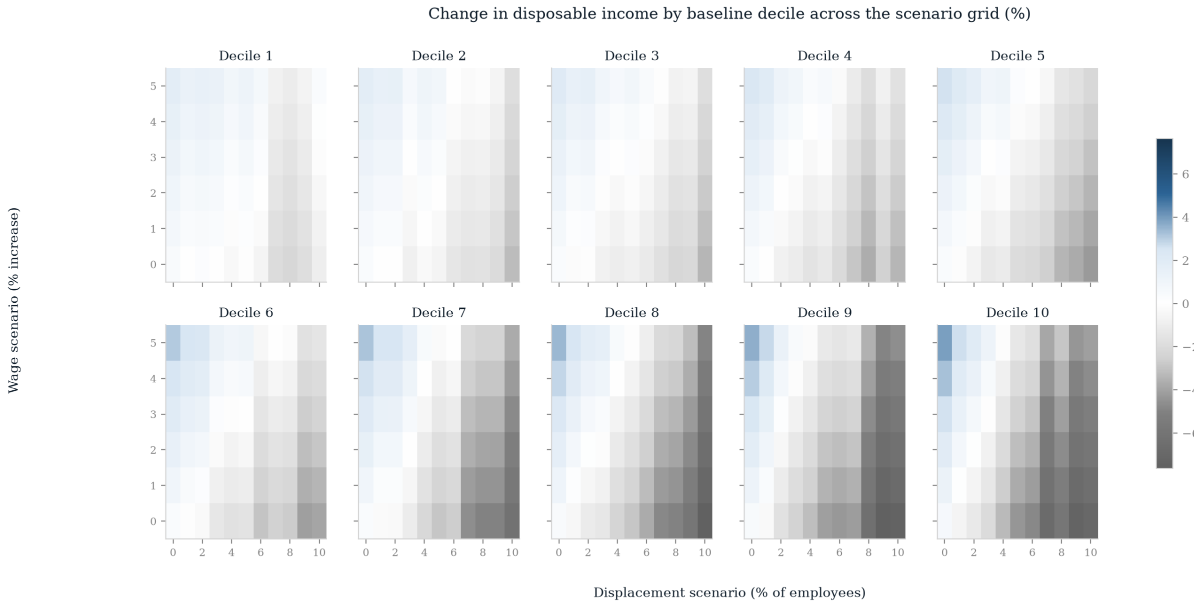

Figure 22: Change in household disposable income across the scenario grid, faceted by baseline income decile. Single draw, seed 0.

Figure 23: Average change in household disposable income across the scenario grid with the capital-return shock switched off. Single draw, seed 0.